Specification book

VFDSVAEEP0002 Basic DID PID Specification

Specifcation Book (SPB)

Aug 30 2023 10:09

VFDSVAEEP0002 _ V1.5.2

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1 Scope

Applicable Document of Part 1

Common Electrical Specification for Supplier

Basic DID Definition Specification

2 Range of Validity

VinFast Vehicles

Author: Daniel Canelopol

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3 Approval Confirmed By

Function	Name	Signature	Date
EEI Chief Engineer	Luca Martinengo / Anirban Gupta	LM/AG	01/02/2022
VinFast Chief Engineer	Paul Newton	PN	01/02/2022

4 General Information

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4.1 Language

This specification is written in English.

4.2 Change History

Version	Chapter	Brief Description	Date	Edited By
1.0.0	All	1. Initial Version	*2019.10.14*	Nguyen Hoang Long
1.0.0	All	1. Pdf -> Word Conversion and further update w.r.to C- SUV	2020.01.02	Rasul Nadaf
1.0.1	All	1. Change DID 0xF18A from 5 byte ASCII to 4 byte BCD 2. Changed usage of DID 0xF190 as Mandatory for EMS, BCM & MHU 1. Changed usage of DID 0xF102 (was previously mandatory) 2. Changed parameter example of DID 0xF103, • 7th Byte low nibble is needed • 8th byte marked as ASCII Character "V" to match with	2021.04.25	Rasul Nadaf

		Team centre 1. Changed usage of DID 0xF111 marked for BCM/XGW and MHU 2. Changed usage of DID 0xF112 marked for BCM/XGW and MHU 3. Changed usage of DID 0xF113 marked for BCM/XGW and MHU 4. Changed DID 0xF11A name from PT to Powertrain 5. Changed DID 0xF11B name from TC to Chassis 6. Deleted DID 0xF11F as its not applicable # Changed usage of DID 0xFD01 (was previously mandatory) 7. Changed usage of DID 0xF194(was previously mandatory) 8. Added below New DID as per VF35 architecture • DID 0xF125 ECU ADAS CAN dbc file version • DID 0xF126 ECU Diagnostic CAN dbc file version • DID 0xF127 ECU FA CAN dbc file version • DID 0xF128 ECU FLEXRAY FIBEX file version • DID 0xF129		

		ECU LIN3 LDF file version		
1.1	System Description	1) Specification document name changed to VFDSVAEEP0002. 3) Added below New DID as per architecture: • DID 0xF120 Time Stamp - Global snapshot • DID 0xF121 Odometer - Global snapshot • DID 0xF122 ECU Supply Voltage - Global snapshot	2021.10.22	Daniel Canelopol
		• DID 0xF123 Ignition status - Global snapshot • DID 0xF18C ECUSerialNumberDataIdentifier • DID DD01 Cabin Temperature • DID DD02 Outside Temperature • DID DD03 ECU Temperature • DID DD04 ECU Voltage • DID D00D Vehicle Communication Bus State • DID DD06 Engine status - Global snapshot • DID DD07 GPS Time - Global snapshot • DID DD08 Engine RPM - Global snapshot • DID DD09 Vehicle speed - Global snapshot • DID F18F RegulationXSoftware Identification Number (RxSWIN) • DID D020 Internal Fail Code • DID 4884 Internal Fault Data • DID F800 VIN2		Daniel Canelopol

			• DID F801 VIN3 • DID F802 VIN4 • DID F141 Hardware Number Revision Level • DID F142 Calibration Software Number Revision Level • DID F143 Purchase Part Number Revision Level • DID F144 Application Software Part Number Revision Level • DID F145 Basic Software Part Number Revision Level • DID F146 HMI Software Part Number Revision Level • DID F148 Software Part Number Revision Level 4) Removed DID F18A.		
1.2	System Description	1)	Format of the following DID's has been changed to match Team Centre Part Number – F188, F191,		
		2) 3)	F102, F103, F104, F105, F106. Flash Boot Loader defined for all DID's. CAN signals referenced for Snapshot DID's	2021.11.29	Daniel Canelopol
1.3	System Description	1)	F18C DID – Write once in Bootloader added		
		2)	D005 DID changed to DD05		
		3)	F120 DID name change to Global Time		
		4)	F18F DID – parameter details added		
		5) 6) 7) 8) 9)	D020 DID – parameter details updated 016C DID – new DID added D05B DID – new DID added D06B DID – new DID added F190 DID – Read in Bootloader changed to N/A	2022.01.28	Daniel Canelopol

1.4	3.2 3.14 3.15	1) 2) 3)	DID F188 correctly defined DID F111 no value changed from 00 to FF CRC check updated DIDF112 no value changed from 00 to FF CRC check updated	20/4/2022	Daniel Canelopol & Ashad Qureshi

	3.44 3.31 3.64 3.46 3.48 3.52 3.53 3.54 3.55 3.65 3.18 2.2 3.34 3.35 3.36 3.42 3.5 3.33 3.40 3.41 3.49 3.51 3.63 3.37 3.66 3.67 3.68 3.69 3.70 3.71 3.72 3.73 3.74 3.75 4.2 3.39 3.16 3.18-	1. DID DD05 Motor RPM size increase to 4bytes and note added resolution corrected and right justified 2. DID FD01 Note added 3. DID D06B Usage clarified 4. DID DD07 corrected size to 6 bytes to match F120 5. DD09 typo corrected from 1bit=255 to 1 byte =255km/h 6. Change the requirement of additional VIN 0xF800 storage from M to M/C and a note added 7. Change the requirement of additional VIN 0xF801 storage from M to M/C and a note added 8. Change the requirement of additional VIN 0xF802 storage from M to M/C and a note added 9. Added a 5th VIN 0x0803 for ECU which ties the calibration to a vehicle and note added. 10. DID D05B changed from mandatory to conditional 11. DID 0xF113 DID		

	3.25 3.76 3.77 3.78	length corrected 12. Requirement clarified ref addressing of DID to say all DID support physical and functional addressing 13. DID 0xF110 Removed the ref ECU name added help text for it 1. Correction to the DID length F120 2. DID F121 Note added 3. DD03 ECU temperature clarification note added 4. F190 DID size correctly specified to 24 bytes 5. F194 notes added 6. DD01 Use case clarified 7. DD02 Use case clarified 8. D00D DID clarified and notes added. 9. DID4884 clarified and type changed from hex to packeted 10. DID 0x016C Changed the DID from unsigned to state encoded and updated table 11. F122 note added for capturing voltage from CAN 12. New DID F16A Current VOD DID for Category 1 ECU 13. New DID F16B		

		Category 1 current and last stored VOD checksum 14. New DID F16C Category 1 Current VOD checksum 15. New DID F16D Category 1 and 2 VOD validation flag 1. 0xEF00 VOD Validation State DID Category 2 ECU 2. 0xEF01VOD Category 2 ECU compared checksum 3. 0xEF02 Last stored VOD for Category 2 ECU 4. 0xEF03 Last stored VOD for Category 1ECU 5. 0xEF04 VOD validation States for Category 1 ECU 6. 0xEF05 Category 2 compare checksum 38) Added reference Document in table 5. 7. 3.39 change the security for writing from the plant to supplier. 8. CDS added to the title		
	3.79 3.43 3.3 3.66	1. CAN DB file examples clarified 2. New DID F117 added 3. New DID F147 added 4. New DID 0804 added 5. New DID 0805 added 6. DID D004 Note		

		added 7. F18F clarified as to what it will report 8. Added new DID 0x0222 Valid Application		
1.4 1.4 1.4 1.4 1.4 1.5	3 3.82 3.30 3.83 All 3.40,3.41 3.48 3.37 3.38 3.5	0x016C DID – Updated to align with ODX 0x0179 DID introduced to check ECU control of the contactor 0xF10F DID updated scope Autosar Version number DID added Implementation updated on DIDS DD01 and DD02 requirement clarified DD09 updated F122 vehicle voltage resolution change to match CAN DB F123 Ignition Status DID size changed to match CAN DB F190 Note added regarding writing and the title corrected	24/06/22 12/7/2022 14/07/2022 15/08/2022	Steve Clarke/Ashad Qureshi
1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5 1.5	3.80 3.83 3.84 3.85 3.86 3.77 3.78 3.87 3.88 3.0 All DID 0xF188 Vehicle Manufacturer ECU Software Part Number DID 0xF186 Active Diagnostic Session DID 0xF103 Vehicle Manufacturer ECU Purchase Part Number,	Updated text 0n 0x0805 “Flash” 0x0806 will now be used for “Autosar Version Number” 0x0807 ODX template v DID added 0x0808 EOL template V DID added 0x0809 AS template V DID added DID name changed to F109 from F117 DID name changed to F147 from F149 LIN Node 4 added LIN Node 5 added Info added Requirement to support Functional addressing for all DIDs which are greater than 8 bytes removed Updated DID 0xF18F RXSWIN to comply with ISO14229	23/09/2022 26/09/2022 29/09/2022 3/10/2022 3/10/2022 3/10/2022 11/10/2022 04/11/2022 13/12/2022 13/12/2022 03/01/23	Steve Clarke Ashad Qureshi Ashad Qureshi Ashad Qureshi Ashad Qureshi Ashad Qureshi Steve Clarke A Qureshi A Qureshi Steve Clarke

	DID 0xF104 Vehicle Manufacturer ECU Application Software Part Number DID Standard for service ID 0x22, DID 0xD00D Vehicle Network Communication Bus State	2020 where DID is encoded all ASCII F188 usage redefined F104, F105 Usage redefined Added section 4 DID Standard for service ID 0x22 plus a short working description DID 0xD00D is now strikethrough – not to be used		
1.5.2	DID 0xDB11 DID 0xCB01 -> 0xCBFF Terms and definitions Additional document	Add new DID to check Active Bank in Dual Bank Add new DID for SecureCAN (as per CyberSecurity spec version V2.0) Add abbreviation for "CS" and "SW" Add "CAN_specification" and "CyberSecurity"	29/8/2023	Phí Duy Hải

Table 1: Change History

5 Introduction

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5.1 Coherence to other documents

The “Generic Diagnostic Specification-VFDSVAEEP0001-V1.5” refers to this document.

5.2 Terms and definitions

The following list of terms and definitions are used throughout this document with the given meaning.

All DIDs shall support physical and Functional addressing.

5.2.1 Abbreviations

Abbreviation	meaning
ECU	Electronic Control Unit
DID	Data Identifier
PID	Parameter Identifier
BOOT	Bootloader
Cvt	Convention
C	Conditional
M	Mandatory
O	Optional
U	User Conditional
CS	Cyber Security
SW	software

Table 2 : List of abbreviations Definitions

5.2.2 Obligation of Requirement Wording

The following wording rules are used throughout this document.

Shall

The word "shall" is used to state a binding requirement. Such requirements are externally verifiable by manipulation or observation.

Should

The word "should" is used for preferences or desires in implementation that are not strictly binding.

May

The word "may" is used to denote options that are left to the implementer. Security Level

The number and usage of Security Levels is described in the Generic_Diagnostic_Specification. Basically, the following Security Level exists:

Level	Request Seed	Send Key	Use case

1	0x01	0x02	Service
2	0x03	0x04	Manufacturing
3	0x05	0x06	Variant Coding
4	0x07	0x08	Flashing (ECU reprogramming)

Table 3 : Security Level

Level 0 means no security access is required.

6 System Description

The DIDs detailed in this document are system DIDs to be implemented by ECU suppliers based on use case and to provide useful information such as Part numbers, vehicle status, vehicle environmental condition and snapshot data etc. The suppliers are therefore encouraged to use these DIDs to provide as much detail about their ECUs.

7 DID Standard for service ID 0x22

Data Byte No.	Parameter Name	Cvt	Value
0	Service positive response 0x62	M	
1	DID number upper Byte F1	M	
2	DID number lower Byte 88	M	
3	Data associated with DID	M	1bit to 1 Byte in length (0x01 to 0xFF)
4		C	
5		C	
6		C	
N+1		C	

The structure of all DIDs in this document will be as above where the first byte is byte 0 and will be the ISO14229 Service ID followed by the 2 byte DID value, the 3rd byte will be the first byte of any returned values when using service 0x22 i.e. 0x62 F1 88 [Data byte1]…….[]9 Note attention must be paid to type of DID and Byte to be used

ASCII is an ascii value that represents a physical key stroke Item e.g. ascii 0x30 = keystroke “0”

BCD is binary coded decimal which has following format 1 byte[upper nibble – lower nibble]

So can range from [0 to 9 – 0 to 9] so byte 3 may display 0x99 which = 99 decimal

SED = state encoded DID and has range from 0x00 to 0xFF where each state means something different e.g. 0x00 = OK, 0x01 = Fault, 0xFF means out of range.

A DID my be one type of value e.g. integer, BCD etc. or a mixture for example Part Numbers

“ASCII,ASCII,ASCII,BCD,BCD,BCD,BCD”

Please see detail for each DID listed in the document and cross reference with the latest iso14229 standard.

7.1 DID 0xF186 Active Diagnostic Session

<table>
<tr><td>Service</td><td colspan="3">0x22</td></tr>
<tr><td>DID</td><td colspan="3">0xF186</td></tr>
<tr><td>Short Description</td><td colspan="3">Read Active Diagnostic Session</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td colspan="2">Flash Boot Loader</td></tr>
<tr><td>Mandatory Read only</td><td colspan="2">Read</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td colspan="2">Write</td></tr>
<tr><td>Level 0</td><td colspan="2">N/A</td></tr>
<tr><td>Usage</td><td colspan="3">Mandatory for all CAN ECUs and diagnostic Class III LIN node.</td></tr>
<tr><td>Data</td><td colspan="3">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td>3</td><td>sub-function = [DiagnosticSessionType</td><td></td><td>00 – FF</td></tr>
<tr><td></td><td>defaultSession</td><td>M</td><td>01</td></tr>
<tr><td></td><td>programmingSession</td><td>M</td><td>02</td></tr>
<tr><td></td><td>extendedDiagnosticSession</td><td>M</td><td>03</td></tr>
<tr><td></td><td>coding session</td><td>M/C</td><td>41</td></tr>
<tr><td></td><td>Specific Session</td><td>O</td><td>40</td></tr>
<tr><td></td><td>Specific Session</td><td>O</td><td>50</td></tr>
</table>
</td></tr>
</table>

	]		

Parameter:

- Active Diagnostic Session: 01 = Default Session
  1. = Programming Session
  2. = Extended Diagnostic Session

41 = Coding Session
40 = Specific session Bench Session
50 = Specific session Roller session

Note:

- These are the sub function parameters from Diagnostic Session Control service.
- See also Generic_Diagnostic_Specification chapter “Diagnostic Services”.
- All ECUs need to support sessions 10 01/81, 10 02/82 and 10 03/83 all other sessions are supported as case by case requirements.

7.2 DID 0xF188 Vehicle Manufacturer ECU Software Part Number

Service	0x22				
DID	0xF188				
Short Description	Software Part Number of the ECU defined by the OEM.				
Implementation	**Application**	**Flash Boot Loader**			
	Mandatory Read(physical)	Updated during software download Not readable in Boot Loader			
Security Level	**Read**	**Write**			
	Level 0	N/A			
Usage	Mandatory for all ECUs which are type 2 and type 3				
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**	
		ECU Software Part Number [ ] = [ ]			

	3	ECU Software Number Prefix Byte#0	M	ASCII	
	4	ECU Software Number Prefix Byte#1	M	ASCII	
	5	ECU Software Number Prefix Byte#2	M	ASCII	
	6	ECU Software Number Byte#0	M	BCD	
	7	ECU Software Number Byte#1	M	BCD	
	8	ECU Software Number Byte#2	M	BCD	
	9	ECU Software Number Byte#3	M	BCD	

Parameter:

- ECU Software Part Number from BOM/Teamcenter:

e.g.: SOW12345678

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9
53	4F	57	12	34	56	78

Note:

- The OEM defines Vehicle Manufacture ECU Software Part Number, and it shall be embedded in the ECU Software file.
- This DID is updated during the reprogramming process.
- All Bytes 0x00 means no Vehicle Manufacture ECU Software Part Number is stored.
- Also see DID 0xF148 Vehicle Manufacturer ECU Software Part Number Revision Level.

The ECUs will use this part number to report application and base software where both of these software components are combined. Where they are separate F104 and F105 shall be used.

7.3 DID 0xF18F Regulation-X Software Identification Number (RxSWIN)

Service	0x22
DID	0xF18F

<table>
<tr><td>Short Description</td><td colspan="2">RegulationXSoftware Identification Number (RxSWIN)</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read(physical only)</td><td>Updated during software download Not readable in PBL</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for ECUs that must adhere to RxSWIN Requirements</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td>3 - 13</td><td>RegulationXSoftware Identification Number [ ]</td><td>M</td><td>ASCII</td></tr>
<tr><td>14</td><td>Space</td><td>M</td><td>ASCII</td></tr>
<tr><td>15 -16</td><td>Revision Number</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td></td><td></td><td></td></tr>
</table>
Parameter:
• The values should be aligned left. • The Content will be “ECU Purchase Part Number” then a “Space” then ECU Purchase Part Revision Number

Example
ECU Purchase Part Number = EEP12345678 plus “Space” then “ECU Purchase Part Revision Number” = 01
<table>
<tr><td>Byte 3</td><td>Byte 4</td><td>Byte 5</td><td>Byte 6</td><td>Byte 7</td><td>Byte 8</td><td>Byte 9</td><td>Byte 10</td><td>Byte 11</td><td>Byte 12</td><td>Byte 13</td><td>Byte 14</td><td>Byte 15</td><td>Byte 16</td></tr>
<tr><td>45</td><td>45</td><td>50</td><td>31</td><td>32</td><td>33</td><td>34</td><td>35</td><td>36</td><td>37</td><td>38</td><td>20</td><td>30</td><td>31</td></tr>
</table>
*Unused Bytes should be filled with space (0x20)

• RxSWIN Identification number is a dedicated identifier, representing information about the type approval relevant software of the Electronic Control System contributing to the Regulation Number X type approval relevant characteristics of the vehicle.
• Please refer to ECE-R156 & ISO14229-1 Annex C.

Note 1

• RxSWIN DID will return ECU’s OEM Purchase part numbers related to software. The part numbers returned by an ECU will be relevant only to the ECU. E.g. VCU returns part numbers of VCU only

Note 2
</td></tr>
</table>

	• This DID 0xF18F is all “ASCII” encoded the values are taken from 0xF103 and 0xF143 but must be converted into ASCII. • As Detailed in ISO14229 2020

7.4 DID 0xF18B ECU Manufacturing Date

Service	0x22	
DID	0xF18B	
Short Description	Read ECU Manufacturing Date	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read(physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECUs	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU Manufacturing Date[] = [		
3	ECU Manufacturing Date - Year	M	(00 - 99) BCD
4	ECU Manufacturing Date - Month	M	(01 - 12) BCD
5	ECU Manufacturing Date - Day	M	(01 - 31) BCD
	]		

Parameter:

- Example ECU Manufacturing Date format:

e.g: 2013.01.21

Byte3	Byte4	Byte5

	13 | 01 | 21 **Note:** • Writing the date shall be done at the supplier site. • The ECU Manufacturing Date shows the production date of the ECU. • All values 00 means no ECU Manufacturing Date is stored.

7.5 DID 0xF190 Vehicle Identification Number (VIN)

Service	0x22 /0x2E	
DID	0xF190	
Short Description	Read / Write complete Vehicle Identification Number is (17 Characters) are shown	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read/Write (physical addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	Level1&2
Usage	Mandatory for all ECU's that supports diagnostics	

Data

Data Byte No.	Parameter Name	Cvt	Value
	Vehicle Identification Number [] = [		
3	Vehicle Identification Number High Byte#0	M	ASCII
…			
19	Vehicle Identification Number Low Byte#16	M	ASCII
	]		

Parameter:

- Vehicle Identification Number format: XXXXXXXXXXXXXXXXX ○ e.g.: WBCN1233VVV333212

Note:

	• The VIN write is a onetime write, once written it shall not be possible to overwrite it unless the application software is updated, in which case after the update of the software the VIN will need to be rewritten. • The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process. • The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part. • All Bytes 0x00 means no Vehicle Identification Number is stored. • Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER This signal will only provide the last 7 bytes of the VIN.

7.6 DID 0xF191 Vehicle Manufacturer ECU Hardware Part Number

Service	0x22			
DID	0xF191			
Short Description	Hardware Part Number of the ECU defined by the OEM.			
Implementation	**Application**	**Flash Boot Loader**		
	Mandatory Read (physical addressing)	Read		
Security Level	**Read**	**Write**		
	Level 0	N/A		
Usage	Mandatory			
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**
		ECU Hardware Part Number [ ] = [		
	3	ECU Hardware Number Prefix Byte#0	M	ASCII
	4	ECU Hardware Number Prefix Byte#1	M	ASCII
	5	ECU Hardware Number Prefix Byte#2	M	ASCII
	6	ECU Hardware Number Byte#0	M	BCD

	7	ECU Hardware Number Byte#1	M	BCD
	8	ECU Hardware Number Byte#2	M	BCD
	9	ECU Hardware Number Byte#3	M	BCD
		]		

Parameter:

- ECU Hardware Part Number from BOM/Teamcenter:

e.g.: EEP12345678

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9
45	45	50	12	34	56	78

Note:

- The OEM defines Vehicle Manufacture ECU Hardware Part Number.
- The Hardware Part Number must be written by the Supplier.
- This DID is used to identify a unique combination of ECU Hardware and non-reprogrammable software (e.g. Boot Software).
- If ECU hardware or the Boot Software Version is changed in the ECU delivered to VINFAST, then the Vehicle Manufacturer ECU Hardware Part Number or Revision Level must be changed.
- All Bytes 0x00 means no Vehicle Manufacture ECU Hardware Part Number is stored.
- Also see DID 0xF141 Vehicle Manufacturer ECU Hardware Part Number Revision Level.

7.7 DID 0XF102 Vehicle Manufacturer ECU Calibration Part Number

Service	0x22	
DID	0xF102	
Short Description	Calibration Software Number of the ECU defined by the OEM	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read(physical)	Updated during software download Not readable in Boot Loader

<table>
<tr><td>Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td></td><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for ECUs where Calibration Part Number is to be reported separately</td></tr>
<tr><td></td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Calibration Software Part Number [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU Calibration Number Prefix Byte#0</td><td>M</td><td>ASCII</td></tr>
<tr><td>4</td><td>ECU Calibration Number Prefix Byte#1</td><td>M</td><td>ASCII</td></tr>
<tr><td>5</td><td>ECU Calibration Number Prefix Byte#2</td><td>M</td><td>ASCII</td></tr>
<tr><td>6</td><td>ECU Calibration Number Byte#0</td><td>M</td><td>BCD</td></tr>
<tr><td>7</td><td>ECU Calibration Number Byte#1</td><td>M</td><td>BCD</td></tr>
<tr><td>8</td><td>ECU Calibration Number Byte#2</td><td>M</td><td>BCD</td></tr>
<tr><td>9</td><td>ECU Calibration Number Byte#3</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• ECU Calibration Part Number from BOM/Teamcenter:
e.g.: SOW11223344
<table>
<tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td></tr>
<tr><td>53</td><td>4F</td><td>57</td><td>11</td><td>22</td><td>33</td><td>44</td></tr>
</table>
Note:
• The Calibration Part Number is defined by the OEM, and shall be embedded in the ECU Calibration file.
• This DID is updated during the reprogramming process.
• All Bytes 0x00 means no Vehicle Manufacture ECU Calibration Part Number is stored.
• Also see DID 0xF142 Vehicle Manufacturer ECU Calibration Part Number Revision Level.
</td></tr>
</table>

7.8 DID 0XF101 Boot Loader Version Number

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0xF101</td></tr>
<tr><td>Short Description</td><td colspan="2">Read Boot Loader Version number</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read(physical and functional addressing)</td><td>Read</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for the ECU’s with flash functionality</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>ECU Boot loader version number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU Boot loader version number high byte#0-High nibble Byte</td><td>M</td><td>Unsigned</td></tr>
<tr><td>3</td><td>ECU Boot loader version number high byte#0-Low nibble</td><td>M</td><td>(00-99) BCD</td></tr>
<tr><td>4</td><td>ECU Boot loader version number high byte#1</td><td>M</td><td>(00-99) BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• Example Boot loader version number:

e.g: V12.34

<table>
<tr><td>Byte3 High Nibble</td><td>Byte3 Low Nibble</td><td>Byte4</td></tr>
<tr><td>1</td><td>2</td><td>34</td></tr>
</table>
Note:
• This identifier shall be defined by supplier and written at the supplier side, and shall not be able to be modified during the vehicle life.
• All values 0x00 means no System Supplier Boot Loader Version Number is stored.
</td></tr>
</table>

7.9 DID 0xF103 Vehicle Manufacturer ECU Purchase Part Number

Service	0x22	
DID	0xF103	
Short Description	Vehicle Manufacturer ECU Purchase Part Number	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	Updated during software download Not readable in Boot Loader
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU Part Number [ ] = [		
3	ECU Part Number Prefix Byte#0	M	ASCII
4	ECU Part Number Prefix Byte#1	M	ASCII
5	ECU Part Number Prefix Byte#2	M	ASCII
6	ECU Part Number Byte#0	M	BCD
7	ECU Part Number Byte#1	M	BCD
8	ECU Part Number Byte#2	M	BCD
9	ECU Part Number Byte#3	M	BCD
	]		

Parameter:

- ECU Purchase Part Number from BOM/Team centre:

  e.g. EEP12345678

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9
45	45	50	12	34	56	78

	• **Note:** • The ECU Purchase Part Number is defined by the OEM and must be written at Supplier manufacturing process. • This DID is used to identify a unique combination of ECU Hardware and reprogrammable Software and Calibration. • If ECU Hardware or Software or Calibration are changed in the ECU delivered to VINFAST, then the Vehicle Manufacturer ECU Purchase Part Number or Revision Level must be changed • All Bytes 0x00 means no ECU Purchase Part Number is stored. • Also see DID 0xF143 Vehicle Manufacturer ECU Purchase Part Number Revision Level

7.10 DID 0xF104 Vehicle Manufacturer ECU Application Software Part Number

Service	0x22			
DID	0xF104			
Short Description	Application Software Part Number defined by the OEM			
Implementation	**Application**	**Flash Boot Loader**		
	Mandatory Read(physical and functional addressing)	Updated during software download Not readable in PBL		
Security Level	**Read**	**Write**		
	Level 0	N/A		
Usage	Mandatory for type 1 ECUs with flashing function and whose application software (ASW) packages can be programmed separately from basic software (BSW), else use DID 0xF188			
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**
		ECU Application Software Part Number [ ] =[		
	3	ECU Application Software Number Prefix Byte#0	M	ASCII
	4	ECU Application Software Number	M	ASCII

		Prefix Byte#1		
	5	ECU Application Software Number Prefix Byte#2	M	ASCII
	6	ECU Application Software Number Byte#0	M	BCD
	7	ECU Application Software Number Byte#1	M	BCD
	8	ECU Application Software Number Byte#2	M	BCD
	9	ECU Application Software Number Byte#3	M	BCD
		]		

Parameter:

- Application Software Part Number from BOM/Teamcenter

e.g. SOW12345678

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9
53	4F	57	12	34	56	78

Note:

- The Application Software Number is defined by the OEM, and shall be • embedded in the ECU Application Software file.
- This DID is updated during the reprogramming process.
- All Bytes 0x00 means no Application Software Number is stored.
- Also see DID 0xF144 Vehicle Manufacturer ECU Application Software Part Number Revision Level

7.11 DID 0xF105 Vehicle Manufacturer ECU Basic Software Part Number

Service	0x22
DID	0xF105

<table>
<tr><td>Short Description</td><td colspan="4">Basic Software Part Number defined by the OEM</td></tr>
<tr><td rowspan="2">Implementation</td><td colspan="2">Application</td><td colspan="2">Flash Boot Loader</td></tr>
<tr><td colspan="2">Mandatory Read (physical and functional addressing)</td><td colspan="2">Updated during software download Not readable in PBL</td></tr>
<tr><td rowspan="2">Security Level</td><td colspan="2">Read</td><td colspan="2">Write</td></tr>
<tr><td colspan="2">Level 0</td><td colspan="2">N/A</td></tr>
<tr><td>Usage</td><td colspan="4">Mandatory for type 1 ECUs with flashing function and whose basic software (BSW) packages can be programmed separately from application software (ASW), else use DID 0xF188.</td></tr>
<tr><td rowspan="13">Data</td><td>Data Byte No.</td><td>Parameter Name</td><td>Cvt</td><td>Value</td></tr>
<tr><td></td><td>ECU Basic Software Part Number [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU Basic Software Number Prefix Byte#0</td><td>M</td><td>ASCII</td></tr>
<tr><td>4</td><td>ECU Basic Software Number Prefix Byte#1</td><td>M</td><td>ASCII</td></tr>
<tr><td>5</td><td>ECU Basic Software Number Prefix Byte#2</td><td>M</td><td>ASCII</td></tr>
<tr><td>6</td><td>ECU Basic Software Number Byte#0</td><td>M</td><td>BCD</td></tr>
<tr><td>7</td><td>ECU Basic Software Number Byte#1</td><td>M</td><td>BCD</td></tr>
<tr><td>8</td><td>ECU Basic Software Number Byte#2</td><td>M</td><td>BCD</td></tr>
<tr><td>9</td><td>ECU Basic Software Number Byte#3</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
<tr><td colspan="4"></td></tr>
<tr><td colspan="4">Parameter:
• Basic Software Part Number from BOM/Teamcenter
e.g. SOW12345678</td></tr>
</table>

	<table><tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>Byte8</td><td>Byte9</td></tr><tr><td>53</td><td>4F</td><td>57</td><td>12</td><td>34</td><td>56</td><td>78</td></tr></table> **Note:** • The Basic Software Part Number is defined by the OEM, and shall be • embedded in the ECU Basic Software file. • This DID is updated during the reprogramming process. • All Bytes 0x00 means no Basic Software Part Number is stored. • Also see DID 0xF145 Vehicle Manufacturer ECU Basic Software Part Number Revision Level

7.12 DID 0xF106 Vehicle Manufacturer ECU HMI Software Data Part Number

Service	0x22			
DID	0xF106			
Short Description	HMI Software Part Number defined by the OEM			
Implementation	**Application**	**Flash Boot Loader**		
	Mandatory Read (physical and functional addressing)	Updated during software download Not readable in PBL		
Security Level	**Read**	**Write**		
	Level 0	N/A		
Usage	Mandatory for ECUs with HMI software module and can be reprogrammed.			
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**
		ECU HMI Software Part Number [ ] = [		
	3	ECU HMI Software Number Prefix Byte#0	M	ASCII
	4	ECU HMI Software Number Prefix Byte#1	M	ASCII

	5	ECU HMI Software Number Prefix Byte#2	M	ASCII
	6	ECU HMI Software Number Byte#0	M	BCD
	7	ECU HMI Software Number Byte#1	M	BCD
	8	ECU HMI Software Number Byte#2	M	BCD
	9	ECU HMI Software Number Byte#2	M	BCD
		]		

Parameter:

- HMI Software Part Number from TeamCenter

e.g. SOW12345678

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9
53	4F	57	12	34	56	78

Note:

- This HMI Software Part Number is defined by OEM, and shall be embedded in the ECU HMI file.
- This DID will be updated during the reprogramming process.
- All Bytes 0x00 means no HMI Software Number is stored
- Also see DID 0xF146 Vehicle Manufacturer ECU HMI Software Data Part

Number Revision Level

7.13 DID 0xF107 Programming Fingerprint

Service	0x22 / 0x2E	
DID	0xF107	
Short Description	Programming fingerprint	
Implementation	**Application**	**Flash Boot Loader**

<table>
<tr><td></td><td>Mandatory Read (physical and functional addressing)</td><td>Mandatory Write for Flashable ECU‘s</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>Level 4(programming session)</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for the ECUs with flash functionality</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Write Fingerprint[ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>programming Date [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr>
<tr><td>4</td><td>programming Date [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr>
<tr><td>5</td><td>programming Date [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr>
<tr><td>6</td><td>repairShopCodeOrTesterSerialNumber[byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr>
<tr><td>...</td><td>.....</td><td></td><td></td></tr>
<tr><td>32</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• e.g.: 130128 2W…KL
<table>
<tr><td>Byte3</td><td>Byte4</td><td>Byte5</td><td>Byte6</td><td>Byte7</td><td>...</td><td>Byte 31</td><td>Byte32</td></tr>
<tr><td>13</td><td>01</td><td>28</td><td>2</td><td>W</td><td>...</td><td>K</td><td>L</td></tr>
</table>
•
Note:
• Defined by OEM and written in aftersales.
• Programming Date is BCD code, and default value is 0x00.
• Byte7 to Byte16 represents the Repair Shop Code in ASCII format.
• the following 17 bytes represent the tester Serial number, and the tester serial number is reserved.
• default value is all bytes space(0x20).

0xF131 to 0xF135 are the blocked and used to store the programming fingerprint.
</td></tr>
</table>

	Programming fingerprint written should use DID 0xF107 by the service 0x2E. The data shall be stored at 0xF131 if the fingerprint data is written for the first time. If another data need to be written (also with 0xF107), block 0xF132 should be filled and so on. Reading 0xF107 shall always return that last programming fingerprint that was written. After the last block of 0xF135 is filled and then if all available block is populated, 0xF132 shall be overwritten, then 0xF133 and so on. 0xF131 should be leaved and can't be overwritten. DID F131 to F135 shall be readable as same as DID F107 including the length and format.

7.14 DID 0xF111 Factory Vehicle Order Data VOD Data Identifier

Service	0x22 /0x2E	
DID	0xF111	
Short Description	Read / Write factory Vehicle Order Data VOD	
Implementation	**Application**	**Flash Boot Loader**
	Optional Read / Write	N/A
Security Level	**Read**	**Write**
	Level 0	Level 3
Usage	Mandatory for any ECU category 1 & 2	

Data

Data Byte No.	Parameter Name	Cvt	Value
	Factory Vehicle Order Data VOD [ ] = [		
3	Factory Vehicle Order Data VOD Format Byte	M	Integer
4	Factory Vehicle Order Data VOD Data High Byte#0	M	Mixed
...	....	...	...
n	Factory Vehicle Order Data VOD Data Low Byte#n	M	Mixed
n+1	Factory Vehicle Order Data VOD CRC	M	Integer
	]		

Parameter:
Example Vehicle Order Data VOD:

	• Format Byte = 00000101 = 5 Dec • Length of the Vehicle Order Data VOD Data = 5 Bytes. • The ECU in which factory VOD stored, shall reserve 255 Bytes for factory VOD. **Note:** • This data would be defined by OEM and ECU supplier together, and only can be written once during the manufacturing process at the OEM side. Once written, this can't be changed any more. . • All Bytes FF means no Factory Vehicle Order Data (VOD) is stored • The Factory Vehicle Order Data VOD includes, • the Format Byte 3 will not be included in the CRC calculation, CRC: shall start from byte #4 to n Byte field, which represents the description of the options and topology of the whole vehicle as bit pattern and the CRC is the last Byte. • The Factory Vehicle Order Data VOD is written in two ECUs in the vehicle • For more information please refer the Coding Specification and VOD definition sheet.

7.15 DID 0xF112 Current Vehicle Order Data VOD Data Identifier

Service	0x22 /0x2E	
DID	0xF112	
Short Description	Read / Write Current Vehicle Order Data VOD	
Implementation	**Application**	**Flash Boot Loader**
	Optional Read / Write	N/A
Security Level	**Read**	**Write**
	Level 0	Level 3
Usage	Mandatory for any ECU category 1 & 2	

Data

Data Byte No.	Parameter Name	Cvt	Value
	Current Vehicle Order Data VOD [ ] = [		
3	Current Vehicle Order Data VOD Format Byte	M	Integer

4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed
...	....	...	...
n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed
n+1	Current Vehicle Order Data VOD CRC	M	Integer
	]		

Parameter:
Example Vehicle Order Data VOD:

- Format Byte = 00000101 = 5dec
- Length of the Vehicle Order Data VOD Data = 5 Byte
- The content of this DID is defined and managed by the Project.
- The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD

Note:

- DID would be defined and written by OEM and ECU supplier together.
- All Bytes FF means no Factory Vehicle Order Data VOD is stored.
- the Format Byte 3 will not be included in the CRC calculation, CRC: shall start from byte #4 to n Byte field, which represents the description of the options and topology of the whole vehicle as bit pattern and the CRC is the last Byte.
- Current VOD shall be same as factory VOD if function/features/options of the whole vehicle is never been changed.
- Service shall update this Current VOD should a feature/function/option change in service that is impacted by the coding strategy.
- For more information, please refer the Coding Specification and VOD definition sheet.

7.16 DID 0xF108 Coding Data String CDS Data Identifier

Service	0x22 /0x2E	
DID	0xF108	
Short Description	Read / Write Coding Data String CDS	
Implementation	**Application**	**Flash Boot Loader**
	Optional Read / Write	N/A
Security Level	**Read**	**Write**

	Level 0	Level 3
Usage	Mandatory for the ECUs with CDS Coding functionality	

Data

Data Byte No.	Parameter Name	Cvt	Value
	Coding Data String CDS[ ] = [		
3	CDS Format Byte High Byte#0	M	Integer
4	CDS Format Byte Low Byte#1	M	Integer
5	CDS Short VIN High Byte#0	M	ASCII
...	....	...	....
11	CDS Short VIN Low Byte#6	M	ASCII
12	CDS Data High Byte#0	M	Mixed
...	....	...	....
n	CDS Data Low Byte#n	M	Mixed
n+1	CDS CRC	M	Integer
	]		

Parameter:
Example Coding Data String CDS:

- Format Byte = 00 0010 1000 1100 = 652 Dec
- Length of the Coding Data String = 652 Bytes

Note:

- This data would be defined by OEM together with supplier, and shall be written at OEM site.
- All Bytes 0 means no Coding Data String is stored.
- The Coding Data String includes 10bits CDS format,
- A Short VIN (last 7 characters of the 0xF190 Vehicle Identification Number)
- The n Bytes field, which represent the Coding information as bit pattern and the CRC in the last Byte.
- The Coding Data String field is limited to 1024 Bytes.
- For more information, please refer the Coding Specification.

7.17 DID 0xF10A Coding for CDS Fingerprint

<table>
<tr><td>Service</td><td colspan="2">0x22 /0x2E</td></tr>
<tr><td>DID</td><td colspan="2">0xF10A</td></tr>
<tr><td>Short Description</td><td colspan="2">Coding for CDS fingerprint</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read / Write</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>Level 3</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for the ECUs with Coding functionality.</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Coding for CDS fingerprint[ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>codingDate [byte#1] (YY)</td><td>M</td><td>BCD(00-99)</td></tr>
<tr><td>4</td><td>codingDate [byte#2] (MM)</td><td>M</td><td>BCD(01-12)</td></tr>
<tr><td>5</td><td>codingDate [byte#3] (DD)</td><td>M</td><td>BCD(01-31)</td></tr>
<tr><td>6</td><td>repairShopCodeOrTesterSerialNumber [byte#1] (MSB)</td><td>M</td><td>ASCII</td></tr>
<tr><td>...</td><td>....</td><td>...</td><td>....</td></tr>
<tr><td>32</td><td>repairShopCodeOrTesterSerialNumber [byte#27]</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• Coding for CDS fingerprint format: 130128xxxxxxxxxx
Note:
• Defined and written by OEM.
• All byte 0 means no CDS fingerprint is stored.
• Coding Date is BCD code.
• Byte 7 to Byte16 represents the Repair Shop Code in ASCII format.
</td></tr>
</table>

	• The following17 bytes represent the Tester Serial number, and the Tester serial number is reserved.

7.18 DID 0xF113 Coding for VOD Fingerprint

Service	0x22 /0x2E	
DID	0xF113	
Short Description	Coding for VOD fingerprint	
Implementation	**Application**	**Flash Boot Loader**
	Read / Write	N/A
Security Level	**Read**	**Write**
	Level 0	Level 3
Usage	Mandatory for any ECU category 1 & 2	

Data

Data Byte No.	Parameter Name	Cvt	Value
	Coding for VOD fingerprint[ ] = [		
3	codingDate [byte#1] (YY)	M	BCD(00-99)
4	codingDate [byte#2] (MM)	M	BCD(01-12)
5	codingDate [byte#3] (DD)	M	BCD(01-31)
6	repairShopCodeOrTesterSerialNumber [byte#1] (MSB)	M	ASCII
...	....	...	....
32	repairShopCodeOrTesterSerialNumber [byte#27]	M	ASCII
	]		

Parameter:

	• Coding for VOD fingerprint format: 130128xxxxxxxxxx **Note:** • Defined and written by OEM. • All byte 0 means no VOD fingerprint is stored. • Coding Date is BCD code. • Byte 6 to Byte16 represents the Repair Shop Code in ASCII format. • The following17 bytes represent the Tester Serial number, and the Tester serial number is reserved.

7.19 DID 0xF10C ECU Body CAN dbc file version.

Service	0x22	
DID	0xF10C	
Short Description	ECU body CAN .dbc file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all body CAN ECUs	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU body CAN .dbc version [ ] = [		
3	ECU body CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned
3	ECU body CAN .dbc version -high byte#0- low nibble Byte	M	BCD
4	ECU body CAN .dbc version high byte#1	M	BCD
	]		

Parameter:

	• e.g. 12.34 Byte3 high Nibble | Byte3 low Nibble | Byte4 1 | 2 | 34 • **Note:** • Defined by OEM, but written at the supplier side. • All byte 0 means no ECU DBC file version is stored. • All ECU's connected on Body CAN should support this DID.

7.20 DID 0xF11A ECU Powertrain CAN dbc file version

Service	0x22	
DID	0xF11A	
Short Description	ECU PT CAN .dbc file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all Powertrain CAN ECUs	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU Powertrain CAN .dbc version [ ] = [		
3	ECU Powertrain CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned
3	ECU Powertrain CAN .dbc version -high byte#0-low nibble Byte	M	BCD
4	ECU Powertrain CAN .dbc version high byte#1	M	BCD
	]		

	Parameter: • e.g. 12.34 Byte3 high Nibble | Byte3 low Nibble | Byte4 1 | 2 | 34 • **Note:** • Defined by OEM, but written at the supplier side. • All byte 0 means no ECU DBC file version is stored. • All Powertrain CAN ECU should support this DID.

7.21 DID 0xF11B ECU Chassis CAN dbc file version

Service	0x22	
DID	0xF11B	
Short Description	ECU Chassis CAN .dbc file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all Chassis CAN ECUs	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU Chassis CAN .dbc version [ ] = [		
3	ECU Chassis CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned
3	ECU Chassis CAN .dbc version -high byte#0- low nibble Byte	M	BCD
4	ECU Chassis CAN .dbc version high byte#1	M	BCD

	]			

Parameter:

- e.g. 12.34

Byte3 high Nibble	Byte3 low Nibble	Byte4
1	2	34

- **Note:**
- Defined by OEM, but written at the supplier side.
- All byte 0 means no ECU DBC file version is stored.
- All Chassis CAN ECU should support this DID.

7.22 DID 0xF10E ECU Infotainment CAN dbc file version

Service	0x22	
DID	0xF10E	
Short Description	ECU Infotainment CAN .dbc file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all Infotainment CAN ECUs	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU Infotainment CAN .dbc version [ ] = [		
3	ECU Infotainment CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned
3	ECU Infotainment CAN .dbc version -high byte#0-low nibble Byte	M	BCD

	4	ECU Infotainment CAN .dbc version high byte#1	M	BCD
		]		

Parameter's

- e.g. 12.34

Byte3 high Nibble	Byte3 low Nibble	Byte4
1	2	34

- **Note:**
- Defined by OEM, but written at the supplier side.
- All byte 0 means no ECU DBC file version is stored.
- All Infotainment CAN ECUs should support this DID.

7.23 DID 0xF125 ECU ADAS CAN dbc file version

Service	0x22	
DID	0xF125	
Short Description	ECU connected to ADAS CAN .dbc file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for ECU's on ADAS CAN	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU ADAS CAN .dbc version [ ] = [		
3	ECU ADAS CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned

	3	ECU ADAS CAN .dbc version -high byte#0- low nibble Byte	M	BCD
	4	ECU ADAS CAN .dbc version high byte#1	M	BCD
		]		

Parameter:

- e.g. 12.34

Byte3 high Nibble	Byte3 low Nibble	Byte4
1	2	34

- **Note:**
- Defined by OEM, but written at the supplier side.
- All byte 0 means no Diagnostic DBC file version is stored.
- All ADAS CAN ECU should support this DID.

7.24 DID 0xF126 ECU Diagnostic CAN dbc file version

Service	0x22	
DID	0xF126	
Short Description	ECU connected to Diagnostic CAN .dbc file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for ECU's on Diagnostic CAN	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU Diag CAN .dbc version [ ] = [		

3	ECU Diag CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned
3	ECU Diag CAN .dbc version -high byte#0- low nibble Byte	M	BCD
4	ECU Diag CAN .dbc version high byte#1	M	BCD
	]		

Parameter:

- e.g. 12.34

Byte3 high Nibble	Byte3 low Nibble	Byte4
1	2	34

- **Note:**
- Defined by OEM, but written at the supplier side.
- All byte 0 means no Diagnostic DBC file version is stored.
- All Diagnostic CAN ECU should support this DID.

7.25 DID 0xF127 ECU FA CAN dbc file version

Service	0x22	
DID	0xF127	
Short Description	ECU connected to FA CAN .dbc file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for ECU's on FA CAN (VF35 ICE only)	

Data

Data Byte No.	Parameter Name	Cvt	Value

	ECU FA CAN .dbc version [ ] = [		
3	ECU FA CAN .dbc version -high byte#0— High nibble Byte	M	Unsigned
3	ECU FA CAN .dbc version -high byte#0- low nibble Byte	M	BCD
4	ECU FA CAN .dbc version high byte#1	M	BCD
	]		

Parameter:

- e.g. 1234

Byte3 high Nibble	Byte3 low Nibble	Byte4
1	2	34

- **Note:**
- Defined by OEM, but written at the supplier side.
- All byte 0 means no FA DBC file version is stored.
- All FA CAN ECU should support this DID.

7.26 DID 0xF128 ECU FLEXRAY FIBEX file version

Service	0x22	
DID	0xF128	
Short Description	ECU connected to FLEXRAY FIBEX file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for ECU's on FLEXRAY bus (**VF35 ICE only**)	
Data		

Data Byte No.	Parameter Name	Cvt	Value
	ECU Flexray Fibex .dbc version [ ] = [		
3	ECU Flexray Fibex file version -high byte#0— High nibble Byte	M	Unsigned
3	ECU Flexray Fibex file version -high byte#0- low nibble Byte	M	BCD
4	ECU Flexray Fibex file version high byte#1	M	BCD
	]		

Parameter:

- e.g. 1234

Byte3 high Nibble	Byte3 low Nibble	Byte4
1	2	34

- **Note:**
- Defined by OEM, but written at the supplier side.
- All byte 0 means no Flexray Fibex file version is stored.
- All Flexray Fibex FA CAN ECU should support this DID.

7.27 DID 0xF10D ECU LIN1 LDF file version

Service	0x22	
DID	0xF10D	
Short Description	ECU LIN1 .ldf file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all LIN1 ECUs	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU LIN1 .ldf version [ ] = [		
3	ECU LIN1 .ldf version -high byte#0— High nibble Byte	M	Unsigned
3	ECU LIN1 .ldf version -high byte#0- low nibble Byte	M	BCD
4	ECU LIN1 .ldf version high byte#1	M	BCD
	]		

Parameter's

- e.g. 1234

Byte3 high Nibble	Byte3 low Nibble	Byte4
1	2	34

- **Note:**
- Defined by OEM, but written at the supplier side.
- All byte 0 means no ECU LDF file version is stored.
- All LIN1 CAN ECU should support this DID.

7.28 DID 0xF11C ECU LIN2 LDF file version

Service	0x22	
DID	0xF11C	
Short Description	ECU LIN2 .ldf file version	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A

<table>
<tr><td>Usage</td><td>Mandatory for all LIN2 ECUs</td></tr>
<tr><td>Data</td><td>
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>ECU LIN2 .ldf version [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU LIN2 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr>
<tr><td>3</td><td>ECU LIN2 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr>
<tr><td>4</td><td>ECU LIN2 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter’s

e.g. 1234
<table>
<tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr>
<tr><td>1</td><td>2</td><td>34</td></tr>
</table>

Note:
Defined by OEM, but written at the supplier side.
All byte 0 means no ECU LDF file version is stored.
All LIN2 CAN ECU should support this DID.

</td></tr>
</table>

7.29 DID 0xF129 ECU LIN3 LDF file version

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0xF129</td></tr>
<tr><td>Short Description</td><td colspan="2">ECU LIN3 .ldf file version</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td>Security Level</td><td>Read</td><td>Write</td></tr>
</table>

<table>
<tr><td></td><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all LIN3 ECUs</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>ECU LIN3 .ldf version [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU LIN3 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr>
<tr><td>3</td><td>ECU LIN3 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr>
<tr><td>4</td><td>ECU LIN3 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:

e.g. 1234
<table>
<tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr>
<tr><td>1</td><td>2</td><td>34</td></tr>
</table>

Note:
Defined by OEM, but written at the supplier side.
All byte 0 means no BCM LIN1 LDF file version is stored.
All LIN3 ECU should support this DID.

</td></tr>
</table>

7.30 DID 0xF10F Service Date

<table>
<tr><td>Service</td><td colspan="2">0x22 / 0x2E</td></tr>
<tr><td>DID</td><td colspan="2">0xF10F</td></tr>
<tr><td>Short Description</td><td colspan="2">The date of last Service</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read/Write</td><td>N/A</td></tr>
<tr><td>Security Level</td><td>Read</td><td>Write</td></tr>
</table>

<table>
<tr><td></td><td>Level 0</td><td>Level 1</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECU’s which support Software/Calibration/Content Change</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Last Vehicle servicing performed Date[] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Service Date - Year</td><td>M</td><td>(00 - 99) BCD</td></tr>
<tr><td>4</td><td>Service Date - Month</td><td>M</td><td>(01 - 12) BCD</td></tr>
<tr><td>5</td><td>Service Date - Day</td><td>M</td><td>(01 - 31) BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• This date is written by tester tool at the time of a service ID 0x14 (Physical or Functional addressing).
• Any other reasons for writing this DID need to be approved by the VinFast Aftersales Diagnostic and Programming Team
e.g: 2018.01.21
<table>
<tr><td>Byte3</td><td>Byte4</td><td>Byte5</td></tr>
<tr><td>18</td><td>01</td><td>21</td></tr>
</table>
Note:
• The default value is all byte 0x00.
</td></tr>
</table>

7.31 DID 0xFD01 Supplier Specific Part Number Data Identifier

Service	0x22	
DID	0XFD01	
Short Description	Read Supplier specific part number	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional	N/A

	addressing)	
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Optional or User Conditional	

Data

Data Byte No.	Parameter Name	Cvt	Value
	Supplier specific part number [] = [		
3	Supplier Specific Part Number High Byte#0	M	ASCII
…	…..	…	…
3+n-1	Supplier Specific Part Number Low Byte#n	M	ASCII

]

Parameter:

- It is allowed to use a variable String length. See the component diagnostic information. **Note:**
- This DID will capture any supplier specific part number which needs to be read or reported via 0x22 service, if this is supported then the details of its implementation shall be reported in the ODX.

4.32 DID 0xF192 System Supplier ECU Hardware Number

Service	0x22	
DID	0XF192	
Short Description	Read System Supplier hardware part number	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Optional or User Conditional	

Data

Data Byte No.	Parameter Name	Cvt	Value

	Supplier specific hardware number [] = [		
3	Supplier Specific hardware Number High Byte#0	M	ASCII
…	…..	…	…
3+n-1	Supplier Specific hardware Number Low Byte#n	M	ASCII
	]		

Parameter:

- It is allowed to use a variable String length. See also the component diagnostic information. **Note:**
- The system supplier Hardware part number shall be defined and written by the Supplier.

7.32 DID 0xF194 System Supplier ECU Software Number

-Service	0x22	
DID	0XF194	
Short Description	Read Supplier specific Software number	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Optional or User Conditional	

Data

Data Byte No.	Parameter Name	Cvt	Value
	Supplier specific Software number [] = [		
3	Supplier Specific Software Number High Byte#0	M	ASCII
…	…..	…	…
3+n-1	Supplier Specific Software Number Low Byte#n	M	ASCII

		]		

Parameter:

- It is allowed to use a variable String length. See also the component diagnostic information. **Note:**
- The Supplier specific part number shall be stored under this DID details of its intended use shall be provided in the ODX

7.33 DID 0xF110 ECU name data identifier

Service	0x22	
DID	0XF110	
Short Description	Read ECU Name	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic	

Data

Data Byte No.	Parameter Name	Cvt	Value
	ECU Name [] = [		
3	ECU Name High Byte#0	M	ASCII
…	…..	…	…
17	ECU Name Low Byte#n	M	ASCII
	]		

Parameter:

- The max. length of the string is 15 Byte.
- The values should be aligned left.
- Unused Bytes should be filled with space (0x20) **Note:**

	• The ECU will report its name with acronym i.e. BCM = Body Control Module and fill the remaining empty bytes with space ASCII value of space = 0x20Hex so that when its read with a test tool only the ECU name appears BCM

7.34 DID 0xF120 Global Time

Service	0x22	
DID	0XF120	
Short Description	Global Time	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic	

Data

Size (Bytes)	Cvt	Type
6	M	**Unsigned**

Signal name	Start byte	Description	DID length
Time stamp	1. 1	0x F120	48 as defined below
	LSB Byte2	Date of "Year"	8
		Date of "Month"	8
		Date of "Day"	8
		Date of "Hour"	8
		Date of "minute"	8
		Date of "second"	8

	Parameter: • Reference CAN signal message = MHU_DATE_TIME • The values should be aligned left. • Unused Bytes should be filled with space (0x00) **Note:** • This data should capture time from CAN and report it out as part of the global snapshot record. ECU can either support 0xF120 or 0xDD07.

7.35 DID 0xF121 Odometer

Service	0x22	
DID	0XF121	
Short Description	Odometer	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic	
Data	**Size (Bytes)** / **Cvt** / **Type** 3 / M / **Unsigned** **Parameter:** • Reference CAN signal message = IC_MILEAGE • Resolution is 1km/bit. • The values should be aligned left. • Unused Bytes should be filled with space (0x00) **Note:** • The Odometer value is transmitted as 3 bytes with resolution of 1bit/Km with maximum value of 16,777,215 Km once this value has reached the ODO meter value shall not increment any further.	

7.36 DID 0xF122 Vehicle Supply Voltage

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0XF122</td></tr>
<tr><td>Short Description</td><td colspan="2">Vehicle Supply Voltage</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECU’s which support diagnostic</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Size (Bits)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>14</td><td>M</td><td>Unsigned</td></tr>
</table>
Parameter:
• Reference CAN signal message = BCM_VOLTAGE
• Resolution is 0.000975625V/bit • The values should be aligned left.
• Unused Bytes should be filled with space (0x00) Note:
• This Voltage value will be on CAN under BCM_VOLTAGE which can be captured into this DID and reported out as a snapshot
</td></tr>
</table>

7.37 DID 0xF123 Ignition status

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0XF123</td></tr>
<tr><td>Short Description</td><td colspan="2">Ignition status</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
</table>

<table>
<tr><td>Usage</td><td>Mandatory for all ECU’s which support diagnostic</td></tr>
<tr><td>Data</td><td>
<table>
<tr><th>Size (Bits)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>3</td><td>M</td><td>Unsigned</td></tr>
</table>
Parameter:

• 0x00=OFF
• 0x01=ACC
• 0x02=ON
• 0x03=START
• 0x04~0x06=reserved
• 0x07=invalid

• Reference CAN signal message = BCM_CLAMP_STAT • The values should be aligned left.
• Unused Bytes should be filled with space (0x00)

Note:

• This DID will indicate the status of the ignition
</td></tr>
</table>

7.38 DID 0xF18C ECU SerialNumberDataIdentifier

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0XF18C</td></tr>
<tr><td>Short Description</td><td colspan="2">ECUSerialNumberDataIdentifier</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>Read (plant)/ Write once only (supplier)</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>NA for plant</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECU’s which support diagnostic</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>16</td><td>M</td><td>ASCII</td></tr>
</table>
</td></tr>
</table>

	Parameter: • The values should be aligned right. • Unused Bytes should be filled with space (0x00) **Note:** • The ECU Serial number should be written **ONCE** only by the Supplier at their production facilities.

7.39 DID 0xDD01 Cabin temperature

Service	0x22	
DID	0XDD01	
Short Description	Cabin temperature	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic	
Data		

Size (Bytes)	Cvt	Type
1	M	**Unsigned**

Parameter:

- Reference CAN signal message = CCU_INTERIORTEMP • Supported based on ECU location.
- The values should be aligned left.
- Unused Bytes should be filled with space (0x00)

Length = 8 Bits
Unit = Degree C
Value Type = Unsigned
Initial Value = 0
Factor = 0.5
Offset = -40
Min value = -40
Max value = +85 **Note:**

	• This will be required to be captured by ECUs which does not have an on board sensor to measure its surrounding temperature and capturing this temperature from the CAN is a lose substitute for the actual temperature and it is based on the location of the ECU, for example an ECU housed in the *trunk (*unless the trunk is a part of the cabin) or in the engine bay need not support the cabin temperature for these ECUs they will need to capture the outside temperature defined in 0xDD02 DID

7.40 DID 0xDD02 Outside temperature

Service	0x22	
DID	0XDD02	
Short Description	Outside temperature	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic which are located on outside of the vehicle.	

Data

Size (Bytes)	Cvt	Type
1	M	**Unsigned**

Parameter:

- Reference CAN signal message = TEMP_EX_UNFILT • Supported based on ECU location • The values should be aligned left.
- Unused Bytes should be filled with space (0x00)

Length = 8 Bits
Unit = Degree C
Value Type = Unsigned
Initial Value = 0
Factor = 0.5
Offset = -40
Min value = -40
Max value = +85 **Note:**
This will be required to be captured by ECUs which does not have an on board

	sensor to measure its surrounding temperature and capturing this temperature from the CAN is a lose substitute for the actual temperature and it is based on the location of the ECU. For example, an ECU which is not located outside of vehicle cabin does not need to support this DID.

7.41 DID 0xDD03 ECU temperature

Service	0x22	
DID	0XDD03	
Short Description	ECU temperature	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic	
Data		

Size (Bytes)	Cvt	Type
1	M	**Unsigned**

Parameter:

- The values should be aligned left.
- Unused Bytes should be filled with space (0x00) **Note:**
- The ECU Temperature is required to be recorded to be reported as part of the local ECU snapshot where there is no hardware to support this a cabin or outside temperature reading shall be provided to accompany any failure as a substitute. The main reason for this to exclude the environment under which the ECU was operating as a contribution factor.

7.42 DID 0xDD04 ECU Voltage

Service	0x22
DID	0XDD04

<table>
<tr><td>Short Description</td><td colspan="2">ECU Voltage</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECU’s which support diagnostic</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>1</td><td>M</td><td>Unsigned</td></tr>
</table>
Parameter:

Resolution is 0.1V/bit
The values should be aligned left.
Unused Bytes should be filled with space (0x00) Note:
This DID will capture the battery supply voltage at its input pins which will be helpful to compare against the vehicle supply voltage being transmitted on CAN to determine if there is a supply issue causing the fault.
Where the local supply voltage cannot be captured due to hardware limitation, then the ECU shall capture the vehicle supply voltage and associate this with the DTC

</td></tr>
</table>

7.43 DID 0xDD05 Motor RPM

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0XDD05</td></tr>
<tr><td>Short Description</td><td colspan="2">Motor RPM</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECU’s which support diagnostic</td></tr>
</table>

Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>**4**</td><td>O/M</td><td>**Unsigned**</td></tr></table> **Parameter:** • **Bytes 1-2**: Reference CAN signal message = EDS_SPD_FR • **Byte 3-4**: Reference CAN signal message = EDS_SPD_RR • Supported only on BEV vehicles. • The values should be aligned Right. • Unused Bytes should be filled with space (0x00) **Note:** • Where a single motor is fitted only the axle that it is attached to the relevant bytes shall report the motor RPM. i.e. 1-2 for the front and 3-4 for the rear. The Motor RPM shall be directly mapped from the CAN Resolution is 1rpm/bit and offset is -20000 RPM, with range value from -20000 RPM to 20000 RPM.

7.44 DID 0xDD06 Engine status

Service	0x22	
DID	0XDD06	
Short Description	Engine status	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic **(ICE only)**	
Data	<table><tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr><tr><td>1</td><td>O/M</td><td>**Bit Mapped**</td></tr></table>**Parameter:** • Supported only on ICE vehicles. • The values should be aligned left.	

	Note: • The ECU name shall be defined by the OEM, but should be written by the Supplier.

7.45 DID 0xDD07 GPS Time

Service	0x22	
DID	0XDD07	
Short Description	GPS Time	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic	

Data

Size (Bytes)	Cvt	Type
6	O/M	**Unsigned**

Signal name	Start byte	Description	DID length
Time stamp	1. 1	0x DD07	48 as defined below
		Date of "Year"	8
		Date of "Month"	8
		Date of "Day"	8
		Date of "Hour"	8
		Date of "minute"	8
		Date of "second"	8

	Parameter: • The length of the DID is 6 Byte the year will be 1 byte i.e. 22 as opposed 2022. • Reference CAN signal message = MHU_DATE_TIME • The values should be aligned left. • Unused Bytes should be filled with space (0x00) **Note:** • This data should capture time from CAN and report it out as part of the global snapshot record. ECU can either support 0xF120 or 0xDD07 depending upon where they get the signal from.

7.46 DID 0xDD08 Engine RPM

Service	0x22	
DID	0XDD08	
Short Description	Engine RPM	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic **(ICE only)**	
Data	**Size (Bytes)** / **Cvt** / **Type** 1 / O/M / **Unsigned** **Parameter:** • Supported only on ICE vehicles. • The values should be aligned left. • Unused Bytes should be filled with space (0x00) **Note:** • The ECU name shall be defined by the OEM, but should be written by the Supplier. • Resolution is 50/bit = 12750 with max value of 10000RPM.	

7.47 DID 0xDD09 Vehicle speed

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0XDD09</td></tr>
<tr><td>Short Description</td><td colspan="2">Vehicle Speed</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECU’s which support diagnostic only if this is required by the ECU</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Size (Bits)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>13</td><td>M/C</td><td>Unsigned</td></tr>
</table>
Parameter:
• Reference CAN signal message = VEHICLESPD • The values should be aligned left.
• Unused Bytes should be filled with space (0x00) Note:
• Resolution is 13/bits = 460km but is limited 300km/h • Factor = 0.05625.
</td></tr>
</table>

7.48 DID 0xD00D Vehicle Network Communication Bus State

<table>
<tr><td><s>Service</s></td><td colspan="2"><s>0x22</s></td></tr>
<tr><td><s>DID</s></td><td colspan="2"><s>0XD00D</s></td></tr>
<tr><td><s>Short Description</s></td><td colspan="2"><s>Vehicle Network Communication Bus State</s></td></tr>
<tr><td rowspan="2"><s>Implementation</s></td><td><s>Application</s></td><td><s>Flash Boot Loader</s></td></tr>
<tr><td><s>Mandatory Read (physical and functional addressing)</s></td><td><s>N/A</s></td></tr>
</table>

<table>
<tr><td>Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td></td><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Optional for all ECU’s which support diagnostic</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>5</td><td>O</td><td>Packeted</td></tr>
</table>
<table>
<tr><th>State 0-3Bits SED</th><th>CAN</th><th>Fault State Bits 4-7 SED</th><th>Faults</th><th>YY 1 byte BCD</th><th>MM 1 Byte BCD</th><th>DD 1 Byte BCD</th><th>Min 2 bytes Unsigned INT</th></tr>
<tr><td>0x00</td><td>Initial</td><td>0x00</td><td>No Faults</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0x01</td><td>Body</td><td>0x01</td><td>Plausible</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0x02</td><td>BA</td><td>0x02</td><td>Transmit Error</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0x03</td><td>CH</td><td>0x03</td><td>Receive Error</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0x04</td><td>DIAG</td><td>0x04</td><td>CRC Error</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0x05</td><td>INFOT</td><td>0x05</td><td>Frame Error</td><td></td><td></td><td></td><td></td></tr>
<tr><td>0x06</td><td>PT</td><td>0x06</td><td>Bus Off</td><td></td><td></td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>
No Fault
All entries shall be 00

Example of fault entry one element of the 5 bytes array [CAN SED = 00 Body] [Fault SED= 02 transmit fault] [YY= 22] [MM=04][DD=11] [Min = current time in min] Parameter:

• The values should be aligned left.
• Unused Bytes shall not be reported

Note:

• DID should capture the last 5 states of the Communication Bus/s.
• Please refer to ISO11898 for Error detection and implementation.</td></tr>
</table>

7.49 DID 0xD020 Internal Fail Code

Service	0x22	
DID	0XD020	
Short Description	Internal Fail Code	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional addressing)	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Optional for all ECU's which support diagnostic	

Data

Size (Bytes)	Cvt	Type
105	O	Packeted

1 byte state encoded			**8 bytes Numeric**		**unsigned numeric**
State	**1 byte**		**1/2byte Fault Count max count 8 (8 byte)**	**Fault count**	**Last Time occurrence (96 bytes)**
0000	Failure descri ption	CAN transceiver fault	Number of counts per fault	max count 8	yymmddhhmm
0001	1	name of device failed	Number of counts per fault	max count 8	6
0010	2	name of device failed	Number of counts per fault	max count 8	6
0011	3	name of device failed	Number of counts per fault	max count 8	6
0100	4	name of device failed	Number of counts per fault	max count 8	6
0101	5	name of device failed	Number of counts per fault	max count 8	6

	0110	6	name of device failed	Number of counts per fault	max count 8	6
	0111	7	name of device failed	Number of counts per fault	max count 8	6
	1000	8	name of device failed	Number of counts per fault	max count 8	6
	1001	9	name of device failed	Number of counts per fault	max count 8	6
	1010	10	name of device failed	Number of counts per fault	max count 8	6
	1011	11	name of device failed	Number of counts per fault	max count 8	6
	1100	12	name of device failed	Number of counts per fault	max count 8	6
	1101	13	name of device failed	Number of counts per fault	max count 8	6
	1110	14	name of device failed	Number of counts per fault	max count 8	6
	1111	15	name of device failed	Number of counts per fault	max count 8	6

Parameter:

- The values should be aligned left.
- Unused Bytes should be filled with space (0x00) **Note:**
- The DID data shall be cleared via routine either by the vehicle manufacturer or by the supplier

7.50 DID 0x4884 Internal Fault Data

Service	0x22	
DID	0X4884	
Short Description	Internal Fault Data	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read (physical and functional	N/A

<table>
<tr><td></td><td>addressing)</td><td></td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Optional for all ECU’s which support diagnostic</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>32</td><td>O</td><td>Packeted</td></tr>
</table>
<table>
<tr><td>Faults States SED 4bits</td><td>Type</td><td>Count 4 bits</td><td>Usage of facilities</td><td>Components usage in 2 bytes unsigned Int</td></tr>
<tr><td>00</td><td>No Fault</td><td></td><td>All OK</td><td></td></tr>
<tr><td>01</td><td>Memory fault</td><td></td><td>Kernels</td><td>CPU usage %</td></tr>
<tr><td>02</td><td>CPU</td><td></td><td>Memory</td><td>Memory usage %</td></tr>
<tr><td>03</td><td>Transceiver</td><td></td><td>Network</td><td>Network load %</td></tr>
<tr><td>04</td><td>xx</td><td></td><td>Core temp</td><td>Temp actual value in C^0</td></tr>
<tr><td>05</td><td>xx</td><td></td><td>quiescent current</td><td></td></tr>
<tr><td>06</td><td>xx</td><td></td><td>ECU voltage</td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td></td></tr>
</table>
Parameter:
• The values should be aligned left.
• Unused Bytes should be filled with space (0x00) Note:
• The ECU supplier shall capture all relevant data which they require as part of ECU internal fault which will include capturing things like CPU load Memory usage CPU core temperature etc. to help investigation. It should be appreciated that not every possible fault associated with all of the ECUs can be defined here the above are an example of what can be captured during an event.
</td></tr>
</table>

7.51 DID 0xF800 Vehicle Identification Number 2 (VIN2)

Service	0x22

<table>
<tr><td>DID</td><td colspan="2">0XF800</td></tr>
<tr><td>Short Description</td><td colspan="2">VIN2</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>Level1&2</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory only for all ECU’s which support diagnostic where ECU has a calibration which is specific to a vehicle to prevent the ECU being moved to different vehicle where this occurs a misdetection of VIN should flag in the form of calibration invalid and shall continue to display until it is recalibrated. And the ECU shall check the VIN stored in F803 each ignition cycle</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>M/C</td><td>ASCII</td></tr>
<tr><td>…</td><td></td><td></td><td></td></tr>
<tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>M/C</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• Vehicle Identification Number format: XXXXXXXXXXXXXXXXX ○ e.g.: WBCN1233VVV333212

Note:
• The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process.
• The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part.
• All Bytes 0x00 means no Vehicle Identification Number is stored.
• Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER Note this will

only report the last 7bytes of the VIN</td></tr>
</table>

7.52 DID 0xF801 Vehicle Identification Number 3 (VIN3)

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0XF801</td></tr>
<tr><td>Short Description</td><td colspan="2">VIN3</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>Level1&2</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory only for all ECU’s which support diagnostic where an ECU requires to store VIN to report out as part of DOIP discovery</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>M/C</td><td>ASCII</td></tr>
<tr><td>…</td><td></td><td></td><td></td></tr>
<tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>M/C</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• Vehicle Identification Number format: XXXXXXXXXXXXXXXXX ○ e.g.: WBCN1233VVV333212

Note:
• The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process.
• The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part.
• All Bytes 0x00 means no Vehicle Identification Number is stored.
• Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER
</td></tr>
</table>

7.53 DID 0xF802 Vehicle Identification Number 4 (VIN4)

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0XF802</td></tr>
<tr><td>Short Description</td><td colspan="2">VIN4</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>Level1&2</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory only for all ECU’s which support diagnostic where ECU cannot shared VIN 2 then this VIN should report out VIN as required by SAEJ1979 use case</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>M/C</td><td>ASCII</td></tr>
<tr><td>…</td><td></td><td></td><td></td></tr>
<tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>M/C</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• Vehicle Identification Number format: XXXXXXXXXXXXXXXXX ○ e.g.: WBCN1233VVV333212

Note:
• The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process.
• The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part.
• All Bytes 0x00 means no Vehicle Identification Number is stored.
• Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER
</td></tr>
</table>

7.54 DID 0x0803 Vehicle Identification Number 5 (VIN5)

<table>
<tr><td>Service</td><td colspan="2">0x22/2E</td></tr>
<tr><td>DID</td><td colspan="2">0X0803</td></tr>
<tr><td>Short Description</td><td colspan="2">VIN5</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>Level1&2</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECU’s which support diagnostic in which VIN needs to be stored as part of Calibration which ties the ECU to a vehicle.</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Vehicle Identification Number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Vehicle Identification Number High Byte#0</td><td>• M/C</td><td>ASCII</td></tr>
<tr><td>…</td><td></td><td></td><td></td></tr>
<tr><td>19</td><td>Vehicle Identification Number Low Byte#16</td><td>• M/C</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• Vehicle Identification Number format: XXXXXXXXXXXXXXXXX ○ e.g.: WBCN1233VVV333212
Note:
• The Vehicle Identification Number should be stored as ASCII string with 17 Bytes. The supplier has to fill in all bytes with 0x00 as initial value. The real VIN shall be written in a later process.
• The VIN code would be written into the ECU when this ECU is first fitted to a vehicle, either manufacturing or in service as a replacement part.
• All Bytes 0x00 means no Vehicle Identification Number is stored.
• Reference CAN Signal = BCM_VEHICLE_IDENT_NUMBER
</td></tr>
</table>

- This VIN will only be required to be supported where an ECU has a calibration which has been carried out on vehicle thereby tying the ECU to that specific vehicle if the ECU is moved from the original vehicle to another it should flag a misalignment of calibration requesting calibration be redone with the new vehicle.

7.55 DID 0xF141 Vehicle Manufacturer ECU Hardware Part Number Revision Level

<table>
<tr><td>Service</td><td colspan="4">0x22</td></tr>
<tr><td>DID</td><td colspan="4">0xF141</td></tr>
<tr><td>Short Description</td><td colspan="4">Hardware Number Revision Level of the ECU defined by the OEM.</td></tr>
<tr><td rowspan="2">Implementation</td><td colspan="2">Application</td><td colspan="2">Flash Boot Loader</td></tr>
<tr><td colspan="2">Mandatory Read(physical and functional addressing)</td><td colspan="2">Read</td></tr>
<tr><td rowspan="2">Security Level</td><td colspan="2">Read</td><td colspan="2">Write</td></tr>
<tr><td colspan="2">Level 0</td><td colspan="2">N/A</td></tr>
<tr><td>Usage</td><td colspan="4">Mandatory</td></tr>
<tr><td rowspan="5">Data</td><td>Data Byte No.</td><td>Parameter Name</td><td>Cvt</td><td>Value</td></tr>
<tr><td></td><td>ECU Hardware Part Number Revision Level [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU Hardware Number Revision Level</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
<tr><td colspan="4">Parameter:

• ECU Hardware Part Number Revision Level from BOM/Teamcenter:
• This DID represent the Revision Level of ECU Hardware Part Number found in DID 0xF191

e.g.: EEP12345678-01

<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table></td></tr>
</table>

	Note: • The OEM defines Vehicle Manufacture ECU Hardware Number Revision Level. • The Hardware Number Revision Level must be written by the Supplier. • Byte4 0x00 means no Vehicle Manufacture ECU Hardware Number Revision Level is stored.

7.56 DID 0xF142 Vehicle Manufacturer ECU Calibration Part Number Revision Level

Service	0x22			
DID	0xF142			
Short Description	Calibration Software Number Revision Level of the ECU defined by the OEM			
Implementation	**Application**	**Flash Boot Loader**		
	Mandatory Read	Updated during software download Not readable in Boot Loader		
Security Level	**Read**	**Write**		
	Level 0	N/A		
Usage	Mandatory for ECUs where Calibration Part Number is to be reported separately			
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**
		Calibration Software Part Number Revision Level [ ] = [		
	3	ECU Calibration Software Number Release Level	M	BCD
		]		
	Parameter: • ECU Calibration Software Part Number Release Level from BOM/Teamcenter: • This DID represent the Revision Level of ECU Calibration Software Part Number found in DID 0xF102			

	e.g.: SOW11223344-01 Byte3 01 **Note:** • The Calibration Software Number Revision Level is defined by the OEM, and shall be embedded in the ECU Calibration Software file. • This DID is updated during the reprogramming process. • Byte4 0x00 means no Vehicle Manufacture ECU Calibration Software Number Revision Level is stored.

7.57 DID 0xF143 Vehicle Manufacturer Purchase Part Number Revision Level

Service	0x22			
DID	0xF143			
Short Description	Purchase Part Number Revision Level of the ECU defined by the OEM			
Implementation	**Application**	**Flash Boot Loader**		
	Mandatory Read(physical and functional addressing)	Updated during software download Not readable in Boot Loader		
Security Level	**Read**	**Write**		
	Level 0	N/A		
Usage	Mandatory			
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**
		ECU Part Number Revision Level [ ] = [		
	4	ECU Number Revision Level	M	BCD
		]		
	Parameter: • ECU Purchase Part Number Revision Level from BOM/Teamcenter: • This DID represents the Revision Level of ECU Number found in DID 0xF103			

	e.g. EEP12345678-01 Byte3 01 • **Note:** • The ECU Purchase Part Number Revision Level is defined by the OEM and must be written at Supplier manufacturing process. • Byte4 0x00 means no Vehicle Manufacture ECU Hardware Revision Level is stored.

7.58 DID 0xF144 Vehicle Manufacturer ECU Application Software Part Number Revision Level

Service	0x22			
DID	0xF144			
Short Description	Application Software Part Number Revision Level defined by the OEM			
Implementation	**Application**	**Flash Boot Loader**		
	Read	Updated during software download Not readable in Boot Loader		
Security Level	**Read**	**Write**		
	Level 0	N/A		
Usage	Mandatory for ECUs with flashing function and whose application software (ASW) packages can be programmed separately from basic software (BSW), else use DID 0xF188			
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**
		ECU Application Software Part Number Revision Level [ ] = [		
	3	ECU Application Software Number Revision Level	M	BCD

		]		
	Parameter: • Application Software Part Number Revision Level from BOM/Teamcenter: • This DID represents the Revision Level of Application Software Number found in DID 0xF104 e.g. SOW12345678-01 Byte3 01 **Note:** • The Application Software Number Revision Level is defined by the OEM, and shall be embedded in the ECU Application Software file. • This DID is updated during the reprogramming process. • Byte4 0x00 means no Application Software Number Revision Level is stored.			

7.59 DID 0xF145 Vehicle Manufacturer ECU Basic Software Part Number Revision Level

Service	0x22			
DID	0xF145			
Short Description	Basic Software Part Number Revision Level defined by the OEM			
Implementation	**Application**		**Flash Boot Loader**	
	Read		Updated during software download Not readable in Boot Loader	
Security Level	**Read**		**Write**	
	Level 0		N/A	
Usage	Mandatory for ECUs which with flashing function and whose basic software can be reprogrammed separately			
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**

		ECU Basic Software Part Number Revision Level [ ] = [		
	4	ECU Basic Software Number Revision Level	M	BCD
		]		

Parameter:

- Basic Software Part Number Revision Level from BOM/Teamcenter:
- This DID represents the Revision Level of Basic Software Number found in DID 0xF105

e.g. SOW12345678-01

Byte3
01

Note:

- The Basic Software Part Number Revision Level is defined by the OEM, and shall be embedded in the ECU Basic Software file.
- This DID is updated during the reprogramming process.
- Byte4 0x00 means no Basic Software Part Number Revision Level is stored.

7.60 DID 0xF146 Vehicle Manufacturer ECU HMI Software Part Number Revision Level

Service	0x22	
DID	0xF146	
Short Description	HMI Software Part Number Revision Level defined by the OEM	
Implementation	**Application**	**Flash Boot Loader**
	Read	Updated during software download Not readable in Boot Loader
Security Level	**Read**	**Write**
	Level 0	N/A

<table>
<tr><td>Usage</td><td>Mandatory for ECUs with HMI software module and can be reprogrammed.</td></tr>
<tr><td rowspan="5">Data</td><td colspan="3"><table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>ECU HMI Software Part Number Revision
Level [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU HMI Software Number Revision Level</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table></td></tr>
<tr><td colspan="3">Parameter:

• HMI Software Part Number Revision Level from BOM/Teamcenter.
• This DID represents the Revision Level of HMI Software Number found in DID 0xF106

e.g. SOW12345678-01

<table><tr><td>Byte3</td></tr><tr><td>01</td></tr></table>
Note:

• This HMI Software Part Number Revision Level is defined by the OEM, and shall be embedded in the ECU HMI Software file.

• This DID will be updated during the reprogramming process.

• Byte4 0x00 means no HMI Software Number Revision Level is stored.</td></tr>
</table>

7.61 DID 0xF148 Vehicle Manufacturer ECU Software Part Number

Service	0x22	
DID	0xF148	
Short Description	Software Part Number Revision Level of the ECU defined by the OEM.	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read(physical and functional addressing)	Updated during software download Not readable in Boot Loader

<table>
<tr><td>Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td></td><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECUs.</td></tr>
<tr><td></td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>ECU Software Part Number Revision Level [ ] = [</td><td></td><td></td></tr>
<tr><td>4</td><td>ECU Software Number Revision Level</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
</td></tr>
<tr><td>Data</td><td colspan="2">
Parameter:
• ECU Software Part Number Revision Level from BOM/Teamcenter:
• This DID represents the Revision Level of ECU Software Part Number found in DID 0xF188

e.g.: SOW12345678-01

<table>
<tr><td>Byte3</td></tr>
<tr><td>01</td></tr>
</table>

Note:
• The OEM defines Vehicle Manufacture ECU Software Number Revision Level, and it shall be embedded in the ECU Software file.
• This DID is updated during the reprogramming process.
• Byte4 0x00 means no Vehicle Manufacture ECU Software Revision Number is stored.
</td></tr>
</table>

7.62 DID 0x016C HV Status and Charging Status

Service	0x22
DID	0X016C
Short Description	HV Status and Charging Status

Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for **VCU only**	
Data	**Size (Bytes)** / **Cvt** / **Type**: 1 / M / State Encoded **Parameters:** **4) HV Status: (0000 00xx)** ♣ 0x00= HV OFF ♣ 0x01=HV ON **3) HV Charging Status: (0000 xx00)** ♣ 0x00= Charging OFF ♣ 0x01= Charging ON • The size of each parameter = two bits in length • Unused Bytes parameters 1 and 2 should be filled with (0x00)	

7.63 DID 0xD06B Signed Software Consistency Check

Service	0x22	
DID	0XD06B	
Short Description	Signed Software Consistency Check	
Implementation	**Application**	**Flash Boot Loader**
	Read	Read
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU’s which support diagnostic and is supporting signed software	
Data	**Size (Bytes)** / **Cvt** / **Type**: 1 / M / State encoded	& ok

	Parameters: 00 = Consistency not checked 01 = Consistency checked 02 = Signature doesn't match 03 = Error and detail of the error • The values should be aligned left. • Unused Bytes should be filled with space (0x00) **Note:** • This DID is used as part software consistency check	of cyber security to validate signature along with

7.64 DID 0xD05B Software Status Check

Service	0x22				
DID	0xD05B				
Short Description	Software Part Number Status Check				
Implementation	**Application**	**Flash Boot Loader**			
	Mandatory Read (physical)	Updated during software download Not readable in Boot Loader			
Security Level	**Read**	**Write**			
	Level 0	N/A			
Usage	• Mandatory for all ECUs which support dual bank architecture.				
Data	**Data Byte No.**	**Parameter Name**	**Cvt**	**Value**	
		ECU Software Part Number [ ] = [			
	3	ECU Software Number Prefix Byte#0	• M/C	ASCII	
	4	ECU Software Number Prefix Byte#1	• M/C	ASCII	
	5	ECU Software Number Prefix Byte#2	• M/C	ASCII	

	6	ECU Software Number Byte#0	• M/C	BCD	
	7	ECU Software Number Byte#1	• M/C	BCD	
	8	ECU Software Number Byte#2	• M/C	BCD	
	9	ECU Software Number Byte#3	• M/C	BCD	
	10	ECU Software Number Byte#4	• M/C	State Encoded	

Parameter:

- Byte 4 to Byte 10 = ECU Software Part Number from BOM/Teamcenter
- Byte 11: 1 = accepted; 0 = rejected

e.g.: SOW1234567800

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9	Byte10
53	4F	57	12	34	56	78	00

Note:

- The OEM defines Vehicle Manufacture ECU Software Part Number, and it shall be embedded in the ECU Software file.
- All Bytes 0x00 means no Vehicle Manufacture ECU Software Part Number is stored.

7.65 DID 0x0222 Valid Application

Service	0x22	
DID	0X0222	
Short Description	Valid Application	
Implementation	**Application**	**Flash Boot Loader**
	Read	Read

Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support diagnostic and is supporting signed software	
Data	**Size (Bytes)** / **Cvt** / **Type** 1 M State encoded **Parameters:** 00 = Not Valid 01 = Valid • The values should be aligned left. • Unused Bytes should be filled with space (0x00) **Note:** • This DID is used to store results from Routine 0x0222 Valid	ate Application

7.66 DID 0xF16A Current Vehicle Order Data VOD Data Identifier Cat 1 ECU

Service	0x22 /0x2E	
DID	0xF16A	
Short Description	Read / Write Current Vehicle Order Data VOD	
Implementation	**Application**	**Flash Boot Loader**
	Optional Read / Write	N/A
Security Level	**Read**	**Write**
	Level 0	L3
Usage	Mandatory for any ECU category 1 which supports VOD	
Data	**Data Byte No.** / **Parameter Name** / **Cvt** / **Value** Current Vehicle Order Data VOD [ ] = [	

3	Current Vehicle Order Data VOD Format Byte	M	Integer
4	Current Vehicle Order Data VOD Data High Byte#0	M	Mixed
...	....	...	...
n	Current Vehicle Order Data VOD Data Low Byte#n	M	Mixed
n+1	Current Vehicle Order Data VOD CRC	M	Integer
	]		

Parameter:
Example Vehicle Order Data VOD:

- Format Byte = 00000101 = 5dec
- Length of the Vehicle Order Data VOD Data = 5 Byte
- The content of this DID is defined and managed by the Project.
- The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD

Note:

- DID would be defined and written by OEM and ECU supplier together.
- All Bytes FF means no Vehicle Order Data VOD is stored.
- This DID shall store all of the VOD data to which the ECU subscribes to.
- Service shall update/code this DID upon the requirement of what the ECU needs to supports as part of the software adaptation.
- All parameter values which the ECU supports shall be listed in the ODX of the ECU.
- For more information, please refer the Coding Specification and VOD definition sheet.

7.67 DID 0xF16B Category 1 ECU Current and last stored checksum

Service	0x22	
DID	0XF16B	
Short Description	VOD Checksum	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**

	Level 0	N/A
Usage	Mandatory for all Category 1 ECU's which support VOD	
Data		

Size (Bytes)	Cvt	Type
5	M	**Hex**

Signal name	Start byte	Description	DID length
Time stamp	1. 1	0x F16B	40
		CRC	8
		Checksum	8
		Checksum	8
		Checksum	8
		Checksum	8

Parameter:
i.e. 2ABC571234

Byte1	Byte2	Byte3	Byte4	Byte5
2A	BC	57	12	34

- The values should be aligned left.
- Unused Bytes should be filled with space (0x00) **Note: -**
- This DID shall capture the checksum of the parameter of the value to which the ECU subscribes to and to which the ECU is configured. The values it compares to are received over CAN from the Master and it stores these locally into 0xF112 DID.

7.68 DID 0xF16C Category 1 ECU Current VOD checksum

Service	0x22
DID	0XF16C

Short Description	Current VOD Checksum	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all Category 1 ECU's which support VOD	

Data

Size (Bytes)	Cvt	Type
1	M	**Hex**

Signal name	Start byte	Description	DID length
Time stamp	1. 1	0x F16C	8
		CRC	8

Parameter:

- The values should be aligned left.
- Unused Bytes should be filled with space (0x00) **Note:**
- This DID shall store the checksum of the VOD payload which is stored locally in its DID 0xF112. Which the ECU will use to compare the checksum with the master which it shall receive over CAN.

7.69 DID 0xF16D VOD Validation Flag for Category 1 and 2 ECUs

Service	0x22	
DID	0XF16D	
Short Description	States of ECU VOD validation flag	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**

	Level 0	N/A
Usage	Mandatory for all ECU's which support VOD requirement.	
Data	**Size (Bytes)** / **Cvt** / **Type** 1 / M / State encoded **State:** 00 = Current Configuration Invalid **State:** 01 = Current Configuration Valid. • The values should be aligned left. • Unused Bits shall not be displayed. **Note:** • This DID shall report time a new parameter is required to be set to invalid as soon as V used by all ECUs updating their configuration • The master ECU trigger to transmit the VOD.	the Validation of the current VOD that the ECU has configured to. This shall be supported by both Category 1 and 2 ECU, each absorbed the validation flag shall be OD is validated then the flag shall be set to valid, . after it has validated the VOD internally shall use this as a

7.70 DID 0xEF00 VOD Validation States for a Category 2 ECU

Service	0x22	
DID	0XEF00	
Short Description	States of ECU validation	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support VOD requirement.	
Data	**Size (Bytes)** / **Cvt** / **Type**	

	1	M	State encoded	112 igured DTC U1501-55 DID EF02 last stored configuration Compared and OK 1502-51(Not Programmed) Configuration Parameters U1502-F1 with Configuration Parameters U1502-F0 -94 component or system operating by Software U1502-57 Calibration / Parameter Memory Failure DTC U1502-55 -F2 This DID shall report all of the VOD validation States of Category 2 ECU

State: 00 = Current configuration compared and same as factory
State: 01 = Reading factory configuration DID F111
State: 02 = Reading Current Configuration DID F
State: 03 = Failed Factory VOD not conf
State: 04 = Comparing current configuration
State: 05 = Last Stored configuration
State: 06 = Comparing current and
State: 07 = Current and Stored Configuration
State: 08 = Learning New configuration
State: 09 = Failed to learn Configuration U
State: 0A = Validating Configuration
State: 0B = Conflict Between Values of
State: 0C = Installed Hardware Mismatch
State: 0D = Applying Configuration Update **State:** 0E = Software Execution Error U1502 conditions
State: 0F = Configuration Not Supported
State: 10 = Failed to Store DTC U1502-46
State: 11 = Failed Current VOD not configured
State: 12 = Configure current VOD
State: 13 = Configure Factory VOD
State: 14 = Set Configuration as Valid
State: 15 = Set Configuration as Invalid
State: 16 = Transmit VOD on CAN
State: 17 = Failed to compare DTC U1502

- The values should be aligned left.
- Unused Bits shall not be displayed.

Note:
-

7.71 DID 0xEF01 Category 2 ECU compared checksum

Service	0x22
DID	0XEF01

Short Description	VOD Compared Checksum	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all Category 2 ECU's which support VOD	

Data

Size (Bytes)	Cvt	Type
1	M	**Hex**

Signal name	Start byte	Description	DID length
Time stamp	1. 1	0x EF01	40
		CRC	8

Parameter:

- Reference CAN signal message = VOD2_Checksum • The values should be aligned left.
- Unused Bytes should be filled with space (0x00) **Note:**
- This DID shall be used by the Master and Backup ECUs to store the Validation Checksum to be used by the backup to compare its local version of the VOD payload against what it receives from the Master on CAN. The Master after it has validated the VOD payload shall store it and broadcast it on CAN for other ECU to use and validate their local stored payload which they have acquired from CAN.

7.72 DID 0xEF02 Last Stored Vehicle Order Data VOD Data Cat 2 ECU

Service	0x22 /0x2E	
DID	0xEF02	
Short Description	Last stored Current Vehicle Order Data VOD	
Implementation	**Application**	**Flash Boot Loader**

<table>
<tr><td></td><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>NA</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for Master and Backup ECUs category 2</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Current Vehicle Order Data VOD [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Current Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr>
<tr><td>4</td><td>Current Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr>
<tr><td>...</td><td>....</td><td>...</td><td>...</td></tr>
<tr><td>n</td><td>Current Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr>
<tr><td>n+1</td><td>Current Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
Example Vehicle Order Data VOD:

• Format Byte = 00000101 = 5dec
• Length of the Vehicle Order Data VOD Data = 5 Byte
• The content of this DID is defined and managed by the Project.
• The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD

Note:

• DID would be defined and written by OEM and ECU supplier together.
• All Bytes FF means no Vehicle Order Data VOD is stored.
• This DID shall store the last good configuration of the VOD for Category 2 ECU that has passed all of the validation checks.
• Whenever a new parameter is updated it shall be learned and if it passes the validation check it shall be stored in this DID for comparison when ECU is powered on subsequently with the current DID 0xF112 to check and ensure that no additional parameters have been changed or updated.
• For more information, please refer the Coding Specification and VOD definition sheet.
</td></tr>
</table>

7.73 DID 0xEF03 Last Stored Vehicle Order Data VOD Data Cat 2 ECU

<table>
<tr><td>Service</td><td colspan="2">0x22 /0x2E</td></tr>
<tr><td>DID</td><td colspan="2">0xEF03</td></tr>
<tr><td>Short Description</td><td colspan="2">Last stored Current Vehicle Order Data VOD</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>NA</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for any ECU category 1 which support VOD configuration</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Current Vehicle Order Data VOD [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Current Vehicle Order Data VOD Format Byte</td><td>M</td><td>Integer</td></tr>
<tr><td>4</td><td>Current Vehicle Order Data VOD Data High Byte#0</td><td>M</td><td>Mixed</td></tr>
<tr><td>...</td><td>....</td><td>...</td><td>...</td></tr>
<tr><td>n</td><td>Current Vehicle Order Data VOD Data Low Byte#n</td><td>M</td><td>Mixed</td></tr>
<tr><td>n+1</td><td>Current Vehicle Order Data VOD CRC</td><td>M</td><td>Integer</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
Example Vehicle Order Data VOD:

• Format Byte = 00000101 = 5dec
• Length of the Vehicle Order Data VOD Data = 5 Byte
• The content of this DID is defined and managed by the Project.
• The ECU in which the VOD is stored, shall reserve 255 Bytes for VOD

Note:

• DID would be defined and written by OEM and ECU supplier together.
• All Bytes FF means no Vehicle Order Data VOD is stored.
• This DID shall store the last good configuration of the VOD for Category 1 ECU that has passed all of the validation checks.
</td></tr>
</table>

	• Whenever a new parameter is updated it shall be learned and if it passes the validation check it shall be stored in this DID for comparison when ECU is powered on subsequently with the current DID 0xF112 to check and ensure that no additional parameters have been changed or updated. • For more information, please refer the Coding Specification and VOD definition sheet.

7.74 DID 0xEF04 ECU Validation States for a Category 1

Service	0x22	
DID	0XEF04	
Short Description	States of ECU validation	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support VOD requirement.	

Data

Size (Bytes)	Cvt	Type
1	M	State encoded

State: 00 = Reading VOD Current (DID F112)
State: 01 = Comparing Current & stored configuration
State: 02 = Current and Stored Configuration
DID F16B
State: 03 = Comparing current Configured and last stored configuration
State: 04 = Configuration lost or changed
State: 05 = Learning New configuration
State: 06 = Failed to learn Configuration U
Configuration
State: 07 = Validating Configuration
State: 08 = Conflict Between Values of
State: 09 = Installed Hardware Mismatch
U1052-F1
State: 0A = Applying Configuration

Compared and OK store checksum in

Failed Current VOD and stored does not compare DTC U1052-F2

1052-51 (Not Programmed) Configuration Parameters TC U1052-F0 with Configuration Parameters DTC

U1052-94
-57 DID EF03 (Calibration / Parameter Memory Configuration Compared and OK store Not configured DTCU1052-55

-55

1 ECU

	Update **State:** 0B = Software Execution Error DTC **State:** 0C = Configuration Not Supported by Software DTC U1052 **State:** 0D = Last Stored configuration **State:** 0E = Failed to Store DTC U1052-46 Failure) **State:** 0F = Current configured and Stored checksum in DID F16B **State:** 10 = Failed Current VOD **State:** 11 = Update Current Configured VOD **State:** 12 = Failed Current VOD not configured DTC U1051 **State:** 13 = Current VOD not received on CAN **State:** 14 = Set Configuration as Valid **State:** 15 = Set Configuration as Invalid • The values should be aligned left. • Unused Bits shall not be displayed. **Note:** • This DID shall report all of the VOD validation States of Category	

7.75 DID 0xEF05 Compare Category 2 master and backup ECUs Checksum

Service	0x22	
DID	0XEF05	
Short Description	States of the Master and Backup ECU Payload checksum	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for all ECU's which support VOD requirement.	

Data	**Size (Bytes)**	**Cvt**	**Type**
	1	M	State encoded
	State: 00 = Current VOD compared is NOT **State:** 01 = Current VOD compared is OK. • The values should be aligned left. • Unused Bits shall not be displayed. **Note:** • This DID is required to be stored by the backup be read with service $22 in case there is Master shall be deemed as correct.		OK is used to store the state of the comparison of the checksum between the master and the backup ECU. In case there is a mismatch between the master ECU and Backup ECU to discrepancy the checksum of the

7.76 DID 0XF109 HUD FONT Data Block Part Number

Service	0x22			
DID	0xF109			
Short Description	HUD FONT Data Block Part Number defined by the OEM			
Implementation	**Application**	**Flash Boot Loader**		
	Mandatory Read	Updated during software download Not readable in Boot Loader		
Security Level	**Read**	**Write**		
	Level 0	N/A		
Usage	Mandatory for ECUs Which have large calibration files			
	Data Byte No.	**Parameter Name**	**Cvt**	**Value**
		Calibration Software Part Number [ ] = [		
	3	ECU Calibration Number Prefix Byte#0	M	ASCII
	4	ECU Calibration Number Prefix Byte#1	M	ASCII

	5	ECU Calibration Number Prefix Byte#2	M	ASCII
	6	ECU Calibration Number Byte#0	M	BCD
	7	ECU Calibration Number Byte#1	M	BCD
	8	ECU Calibration Number Byte#2	M	BCD
	9	ECU Calibration Number Byte#3	M	BCD
		]		

Parameter:

- ECU HUD FONT Part Number from BOM/Teamcenter:

e.g.: SOW11223344

Byte3	Byte4	Byte5	Byte6	Byte7	Byte8	Byte9
53	4F	57	11	22	33	44

Note:

- The HUD data block Part Number is defined by the OEM, and shall be embedded in the ECU Calibration file.
- This DID is updated during the reprogramming process.
- All Bytes 0x00 means no Vehicle Manufacture ECU Calibration Part Number is stored.
- Also see DID 0xF149 Vehicle Manufacturer ECU HUD Data block Part Number Revision Level.

7.77 DID 0xF149 HUD FONT Data Block Part Number Revision Level

Service	0x22	
DID	0xF149	
Short Description	HUD FONT Data Block Part Number Revision Level of the ECU defined by the OEM	
Implementation	**Application**	**Flash Boot Loader**
	Mandatory Read	Updated during software download Not readable in Boot Loader
Security Level	**Read**	**Write**
	Level 0	N/A

<table>
<tr><td>Usage</td><td>Mandatory for ECUs Which have large calibration files</td></tr>
<tr><td rowspan="2">Data</td><td>
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Calibration Software Part Number Revision Level [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU Calibration Software Number Release Level</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
</td></tr>
<tr><td>
Parameter:
• ECU HUD FONT part number Release Level from BOM/Teamcenter:
• This DID represents the Revision Level of ECU Calibration Software Part Number found in DID 0xF109

e.g.: SOW11223344-01

<table>
<tr><td>Byte3</td></tr>
<tr><td>01</td></tr>
</table>
Note:
• The HUD FONT Number Revision Level is defined by the OEM, and shall be embedded in the ECU Calibration Software file.
• This DID is updated during the reprogramming process.
• Byte4 0x00 means no Vehicle Manufacture ECU Calibration Software Number Revision Level is stored.
</td></tr>
</table>

7.78 DID 0x0804 Diagnostic Specification Number Applied

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0X0804</td></tr>
<tr><td>Short Description</td><td colspan="2">Read Diagnostic specification number</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td>Security Level</td><td>Read</td><td>Write</td></tr>
</table>

	Level 0	N/A
Usage	Mandatory for All ECUs which use UDS diagnostic requirement	
Data		

Data Byte No.	Parameter Name	Cvt	Value
	Diagnostic Specification Version number [] = [		
3	Diagnostic Specification Version number used Byte#0	M	ASCII
…	…..	…	…
3+n-1	Diagnostic Specification Version number used Low Byte#n	M	ASCII
	]		

Parameter:

- It is allowed to use a variable String length. See also the component diagnostic information.

i.e.Vxxx = V1.4 56 01 2E 04 **Note:**

- The Diagnostic specification version number used in development of the ECU shall be written by the Supplier.

7.79 DID 0x0805 Flash Specification Number Applied

Service	0x22	
DID	0X0805	
Short Description	Read Flash specification number	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for All ECUs which are Flashable	

<table>
<tr><td>Data</td><td>
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Flash Specification Version number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Flash Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr>
<tr><td>…</td><td>…..</td><td>…</td><td>…</td></tr>
<tr><td>3+n-1</td><td>Flash Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• It is allowed to use a variable String length. See also the component diagnostic information.
i.e.Vxxx = V1.4 56 01 2E 04 Note:
• The Flash specification version number used in development of the ECU shall be written by the Supplier.
</td></tr>
</table>

7.80 DID 0x2070 Transport mode

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0X2070</td></tr>
<tr><td>Short Description</td><td colspan="2">Status of Transport Mode</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Only Applicable to the BCM.</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Size (Bytes)</th><th>Cvt</th><th>Type</th></tr>
<tr><td>1</td><td>M</td><td>State encoded</td></tr>
<tr><td colspan="3"></td></tr>
</table>
shall write to
</td></tr>
</table>

	State: 00 = Transport Mode Not Activated **State:** 01 = Transport Mode Activated. **State:** 02 = Failed to activate Transport mode • The values should be aligned left. • Unused Bits shall not be displayed. **Note:** • Transport shall be activated via a routine 9010 and this routine DID 2070 the status of transport mode	

7.81 DID 0x0179 Check ECU Control of HV

Service	0x22	
DID	0x0179	
Short Description	To Check ECU control of HV Contactor	
Implementation	**Application**	**Flash Boot Loader**
	Read	N/A
Security Level	**Read**	**Write**
	Level 0	N/A
Usage	Mandatory for **VCU only**	

Data

Size (Bytes)	Cvt	Type
1	M	State Encoded

Parameters:
1) ECU Control Status: (xx)

- 0x00= ECU has Control
- 0x01= Routine 0xF001 has control

• This check will be helpful to determine who is controlling the contactor ECU or Routine 0xF001

7.82 DID 0x0806 AUTOSAR Version Number Applied

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0X0806</td></tr>
<tr><td>Short Description</td><td colspan="2">Read Autosar Version number</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for All ECUs which use UDS diagnostic requirement and Autosar compliant</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr>
<tr><td>…</td><td>…..</td><td>…</td><td>…</td></tr>
<tr><td>3+n-1 (max 24 bytes)</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• It is allowed to use a variable String length. See also the component diagnostic information.
i.e.Vxxx = V1.4 56 01 2E 04 Note:
• The Autosar version number used in development of the ECU shall be written by the Supplier.
</td></tr>
</table>

7.83 DID 0x0807 ODX Template Version Applied

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0X0807</td></tr>
<tr><td>Short Description</td><td colspan="2">ODX</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for All ECUs which use UDS diagnostic requirement and Autosar compliant</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr>
<tr><td>…</td><td>…..</td><td>…</td><td>…</td></tr>
<tr><td>3+n-1 (Max 24bytes)</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• It is allowed to use a variable String length. See also the component diagnostic information.
i.e.Vxxx = V1.4 56 01 2E 04 Note:
• The ODX template version number used in development of the ECU shall be written by the Supplier.
</td></tr>
</table>

7.84 DID 0x0808 EOL Template Version Applied

<table>
<tr><td>Service</td><td colspan="2">0x22</td></tr>
<tr><td>DID</td><td colspan="2">0X0808</td></tr>
<tr><td>Short Description</td><td colspan="2">End Of Line (EOL)</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for All ECUs which use UDS diagnostic requirement and Autosar compliant</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr>
<tr><td>…</td><td>…..</td><td>…</td><td>…</td></tr>
<tr><td>3+n-1 (Max 24bytes)</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• It is allowed to use a variable String length. See also the component diagnostic information.
i.e.Vxxx = V1.4 56 01 2E 04 Note:
• The ODX template version number used in development of the ECU shall be written by the Supplier.
</td></tr>
</table>

7.85 DID 0x0809 AS Template Version Applied

Service	0x22

<table>
<tr><td>DID</td><td colspan="2">0X0807</td></tr>
<tr><td>Short Description</td><td colspan="2">After Sales (AS) Template</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Read</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for All ECUs which use UDS diagnostic requirement and Autosar compliant</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>Diagnostic Specification Version number [] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>Diagnostic Specification Version number used Byte#0</td><td>M</td><td>ASCII</td></tr>
<tr><td>…</td><td>…..</td><td>…</td><td>…</td></tr>
<tr><td>3+n-1 (Max 24bytes)</td><td>Diagnostic Specification Version number used Low Byte#n</td><td>M</td><td>ASCII</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter:
• It is allowed to use a variable String length. See also the component diagnostic information.
i.e.Vxxx = V1.4 56 01 2E 04 Note:
• The ODX template version number used in development of the ECU shall be written by the Supplier.
</td></tr>
</table>

7.86 DID 0xF13A ECU LIN4 LDF file version

Service	0x22
DID	0xF13A
Short Description	ECU LIN4 .ldf file version

<table>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all LIN4 ECUs</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>ECU LIN4 .ldf version [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU LIN4 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr>
<tr><td>3</td><td>ECU LIN4 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr>
<tr><td>4</td><td>ECU LIN4 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter’s

e.g. 1234
<table>
<tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr>
<tr><td>1</td><td>2</td><td>34</td></tr>
</table>

Note:
Defined by OEM, but written at the supplier side.
All byte 0 means no ECU LDF file version is stored.
All LIN4 CAN ECU should support this DID.

</td></tr>
</table>

7.87 DID 0xF13A ECU LIN5 LDF file version

Service	0x22
DID	0xF13B

<table>
<tr><td>Short Description</td><td colspan="2">ECU LIN5 .ldf file version</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Boot Loader</td></tr>
<tr><td>Mandatory Read (physical and functional addressing)</td><td>N/A</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all LIN4 ECUs</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Value</th></tr>
<tr><td></td><td>ECU LIN5 .ldf version [ ] = [</td><td></td><td></td></tr>
<tr><td>3</td><td>ECU LIN5 .ldf version -high byte#0— High nibble Byte</td><td>M</td><td>Unsigned</td></tr>
<tr><td>3</td><td>ECU LIN5 .ldf version -high byte#0- low nibble Byte</td><td>M</td><td>BCD</td></tr>
<tr><td>4</td><td>ECU LIN5 .ldf version high byte#1</td><td>M</td><td>BCD</td></tr>
<tr><td></td><td>]</td><td></td><td></td></tr>
</table>
Parameter’s

e.g. 1234
<table>
<tr><td>Byte3 high Nibble</td><td>Byte3 low Nibble</td><td>Byte4</td></tr>
<tr><td>1</td><td>2</td><td>34</td></tr>
</table>

Note:
Defined by OEM, but written at the supplier side.
All byte 0 means no ECU LDF file version is stored.
All LIN5 CAN ECU should support this DID.

</td></tr>
</table>

7.88 DID 0xDB11 Check Active Bank (A or B of Dual Bank)

Service	0x22
DID	0xDB11

<table>
<tr><td>Short Description</td><td colspan="2">Check to see which Flash Memory Bank is currently Active</td></tr>
<tr><td rowspan="2">Implementation</td><td>Application</td><td>Flash Bootloader</td></tr>
<tr><td>Read</td><td>Read</td></tr>
<tr><td rowspan="2">Security Level</td><td>Read</td><td>Write</td></tr>
<tr><td>Level 0</td><td>N/A</td></tr>
<tr><td>Usage</td><td colspan="2">Mandatory for all ECUs which supported Dual Bank Flash Memory</td></tr>
<tr><td>Data</td><td colspan="2">
<table>
<tr><th>Data Byte No.</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr>
<tr><td>3</td><td>Active Bank</td><td>O/M</td><td>Unsigned</td><td>State Encoded</td></tr>
</table>
Parameter:
- 0x00: Invalid
- 0x01: Bank A active
- 0x02: Bank B active

Note: <u>*In Dual Bank ECU, the whole Flash memory is divided into two distinct regions, Banks A and B (excluding the bootloader region)</u>
</td></tr>
</table>

7.89 Cybersecurity configuration Data Identifiers

These DIDs help change some Cybersecurity parameters to help fine-tuning Cybersecurity/SecureCAN during development phase.

These DIDs are created based on information of Cybersecurity specification (v2.0) R_2.3.1_08.

Data Type	Data Name	Data Description	Default Value	Max	Min
boolean	*CS_Authentication_Enable	Enable SecureCAN authentication = 1 Disable SecureCAN authentication = 0	1 [0 for disable SecureCAN release 1 for enable SecureCAN release]	1	0
uint16	CS_Tx_FirstAuthentication_Interval	Manual Overwrite Value for Interval between authentication requests before first successful authentication (ms)	100	65535	100
boolean	CS_Tx_CycleTime_TimeOutLimit_E	Manual Overwrite Activation for CS Transmit Timeout Limit (Enable = 1, Disable = 0)	0	1	0
uint16	CS_Tx_CycleTime_TimeOutLimit_O	Manual Overwrite Value for CS Transmit Timeout Limit (ms)	3000	65535	1000
boolean	CS_Tx_WaitForSeed_TimeOutLimit_E	Manual Overwrite Activation for CS Transmit Wait for Seed Timeout Limit (Enable = 1, Disable = 0)	0	1	0
uint16	CS_Tx_WaitForSeed_TimeOutLimit_O	Manual Overwrite Value for CS Transmit Wait for Seed Timeout Limit (ms)	200	65535	5
boolean	CS_Tx_WaitForKey_TimeOutLimit_E	Manual Overwrite Activation for CS Transmit Wait for Key Timeout Limit (Enable = 1, Disable = 0)	0	1	0
uint16	CS_Tx_WaitForKey_TimeOutLimit_O	Manual Overwrite Value for CS Transmit Wait for Key Timeout Limit (ms)	200	65535	5
boolean	CS_Rx_CycleTime_TimeOutLimit_E	Manual Overwrite Activation for CS Receive Timeout Limit (Enable = 1, Disable = 0)	0	1	0
uint16	CS_Rx_CycleTime_TimeOutLimit_O	Manual Overwrite Value for CS Receive Timeout Limit (ms)	3000	65535	1000
boolean	CS_Rx_WaitResponse_E	Manual Overwrite Activation for CS Wait Response Limit (Enable = 1, Disable = 0)	0	1	0
uint16	CS_Rx_WaitResponse_O	Manual Overwrite Value for CS Wait Response Limit (ms)	200	65535	5
boolean	CS_Rx_AccessFailLimit_E	Manual Overwrite Activation for CS Receive Access Fail Limit (Enable = 1, Disable = 0)	0	1	0
uint8	CS_Rx_AccessFailLimit_O	Manual Overwrite Value for CS Receive Access Fail Limit	5	255	1
boolean	CS_Tx_AuthFailLimit_E	Manual Overwrite Activation for CS Transmit Authentication Fail Limit (Enable = 1, Disable = 0)	0	1	0
uint8	CS_Tx_AuthFailLimit_O	Manual Overwrite Value for CS Transmit Authentication Fail Limit	5	255	1
boolean	CS_<Self>_<ECUB>_Auth_E	Manual Overwrite for Activation of authentication between <Self> and <ECUB> (Enable = 1, Disable = 0)	0	1	0
boolean	CS_SECCAN_<MSGID>_E	Manual Overwrite Activation for Secret portion verification of <MSGID> (Enable = 1, Disable = 0)	0	1	0

7.89.1 **DID 0xCB01 Enable/Disable SecureCAN authentication**

Service	0x22 / 0x2E	
DID	0xCB01	
Short Description	Enable/Disable SecureCAN authentication.	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN	

Data	<table><tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr><tr><td>1</td><td>CS_Authentication_Enable</td><td>O/M</td><td>Boolean</td><td>State Encoded</td></tr></table> **Parameter:** 0x00 - Disable 0x01 - Enable **Note:** If “CS_Authentication_Enable” DID is set to 0, then both ECU authentication and secret portion check will be disabled.

7.89.2 **DID 0xCB02 Overwrite value for interval between authentication requests**

Service	0x22 / 0x2E	
DID	0xCB02	
Short Description	Manual Overwrite Value for Interval between authentication requests before first successful authentication (ms)	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN	

<table>
<tr><td>Data</td><td>
<table>
<tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr>
<tr><td>2</td><td>CS_Tx_First Authentication_Interval</td><td>O/M</td><td>Unsigned</td><td>0x0064 - 0xFFFF</td></tr>
</table>
Parameter:
Min - 0x0064 (decimal: 100)
Max - 0xFFFF (decimal: 65535)
Default - 0x0064 (decimal: 100)

Note: This value is used to define the timeout interval between authentication requests (27 01) for the Sender ECU during power on before the first successful authentication
</td></tr>
</table>

7.89.3 **DID 0xCB03 Overwrite activation for Transmit Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB03	
Short Description	Manual Overwrite Activation for CS Transmit Timeout Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	This is an optional parameter. It is required to be implemented in SecureCAN ECUs which want to maintain two values, one from DID and one coded in SW.	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
1	CS_Tx_CycleTime_TimeOutLimit_E	O/M	Boolean	State Encoded

Parameter:
0x00 - Disable
0x01 - Enable

Note: This is parameters enables the overwrite for CS TransmitTimeout Limit. This needs to be done only if the supplier maintains two configurations, DID configuration and software coded configuration.

7.89.4 **DID 0xCB04 Overwrite value for Transmit Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB04	
Short Description	Manual Overwrite Value for CS Transmit Timeout Limit (ms)	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN and DID 0xCB03	

<table>
<tr><td>Data</td><td>
<table>
<tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr>
<tr><td>2</td><td>CS_Tx_CycleTime_TimeOutLimit_O</td><td>O/M</td><td>Unsigned</td><td>0x03E8 - 0xFFFF</td></tr>
</table>
Parameter:
Min - 0x03E8 (decimal: 1000)
Max - 0xFFFF (decimal: 65535)
Default - 0x0BB8 (decimal: 3000)

Note: This value is used to define the authentication cycle time or the time between one authentication and the next authentication cycle (Sender ECU)
</td></tr>
</table>

7.89.5 **DID 0xCB05 Overwrite activation for Transmit Wait for Seed Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB05	
Short Description	Manual Overwrite Activation for CS Transmit Wait for Seed Timeout Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	This is an optional parameter. It is required to be implemented in SecureCAN ECUs which want to maintain two values, one from DID and one coded in SW.	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
1	CS_Tx_WaitForSeed_TimeOutLimit_E	O/M	Boolean	State Encoded

Parameter:
0x00 - Disable
0x01 - Enable

Note: This is parameters enables the overwrite for CS Transmit Wait for Seed timeout limit. This needs to be done only if the supplier maintains two configurations, DID config and software coded config

7.89.6 **DID 0xCB06 Overwrite value for Transmit Wait for Seed Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB06	
Short Description	Manual Overwrite Value for CS Transmit Wait for Seed Timeout Limit (ms)	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN and DID 0xCB05	

Data	<table><tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr><tr><td>2</td><td>CS_Tx_WaitForSeed_TimeOutLimit_O</td><td>O/M</td><td>Unsigned</td><td>0x0005 - 0xFFFF</td></tr></table> **Parameter:** Min - 0x0005 (decimal: 5) Max - 0xFFFF (decimal: 65535) Default - 0x00C8 (decimal: 200) **Note:** This value defines the time that Sender ECU needs to wait for the SEED (67 01) message from the Receiver ECU after sending SEED REQUEST (27 01)

7.89.7 **DID 0xCB07 Overwrite activation for Transmit Wait for Key Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB07	
Short Description	Manual Overwrite Activation for CS Transmit Wait for Key Timeout Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	This is an optional parameter. It is required to be implemented in SecureCAN ECUs which want to maintain two values, one from DID and one coded in SW.	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
1	CS_Tx_WaitForKey_TimeOutLimit_E	O/M	Boolean	State Encoded

Parameter:
0x00 - Disable
0x01 - Enable

Note: This is parameters enables the overwrite for CS Transmit Wait for Key timeout limit. This needs to be done only if the supplier maintains two configurations, DID config and software coded config

7.89.8 **DID 0xCB08 Overwrite value for Transmit Wait for Key Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB08	
Short Description	Manual Overwrite Value for CS Transmit Wait for Key Timeout Limit (ms)	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN and DID 0xCB07	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
2	CS_Tx_WaitForKey_TimeOutLimit_O	O/M	Unsigned	0x0005 - 0xFFFF

Parameter:
Min - 0x0005 (decimal: 5)
Max - 0xFFFF (decimal: 65535)
Default - 0x00C8 (decimal: 200)

Note: This value defines the time that Sender ECU needs to wait for the KEY K2 (67 02) message from the Receiver ECU after sending KEY K1 (27 02)

7.89.9 **DID 0xCB09 Overwrite activation for Receive Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB09	
Short Description	Manual Overwrite Activation for CS Receive Timeout Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	This is an optional parameter. It is required to be implemented in SecureCAN ECUs which want to maintain two values, one from DID and one coded in SW.	

Data	<table><tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr><tr><td>1</td><td>CS_Rx_CycleTime_TimeOutLimit_E</td><td>O/M</td><td>Boolean</td><td>State Encoded</td></tr></table> **Parameter:** 0x00 - Disable 0x01 - Enable **Note:** This is parameters enables the overwrite for CS Receive Timeout Limit. This needs to be done only if the supplier maintains two configurations, DID config and software coded config

7.89.10 **DID 0xCB10 Overwrite value for Receive Timeout Limit**

Service	0x22 / 0x2E	
DID	0xCB10	
Short Description	Manual Overwrite Value for CS Receive Timeout Limit (ms)	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN and DID 0xCB09	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
2	CS_Rx_CycleTime_TimeOutLimit_O	O/M	Unsigned	0x03E8 - 0xFFFF

Parameter:
Min - 0x03E8 (decimal: 1000)
Max - 0xFFFF (decimal: 65535)
Default - 0x0BB8 (decimal: 3000)

Note: This value is used to define the authentication cycle time or the time between one authentication and the next authentication cycle (Receiver ECU).

7.89.11 **DID 0xCB11 Overwrite activation for Wait Response Limit**

Service	0x22 / 0x2E	
DID	0xCB11	
Short Description	Manual Overwrite Activation for CS WaitResponse Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	This is an optional parameter. It is required to be implemented in SecureCAN ECUs which want to maintain two values, one from DID and one coded in SW.	

Data	<table><tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr><tr><td>1</td><td>CS_Rx_WaitResponse_E</td><td>O/M</td><td>Boolean</td><td>State Encoded</td></tr></table> **Parameter:** 0x00 - Disable 0x01 - Enable **Note:** This is parameters enables the overwrite for CS Wait for Response limit. This needs to be done only if the supplier maintains two configurations, DID config and software coded config

7.89.12 **DID 0xCB12 Overwrite value for Wait Response Limit**

Service	0x22 / 0x2E	
DID	0xCB12	
Short Description	Manual Overwrite Value for CS Wait Response Limit (ms)	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN and DID 0xCB11	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
2	CS_Rx_WaitResponse_O	O/M	Unsigned	0x0005 - 0xFFFF

Parameter:
Min - 0x0005 (decimal: 5)
Max - 0xFFFF (decimal: 65535)
Default - 0x00C8 (decimal: 200)

Note: Optional parameter which defines the time the ECU needs to wait before responding to a request from Sender ECU. This is used to fine tune the response speed.

7.89.13 **DID 0xCB13 Overwrite activation for Receive Access Fail Limit**

Service	0x22 / 0x2E	
DID	0xCB13	
Short Description	Manual Overwrite Activation for CS Receive Access Fail Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	This is an optional parameter. It is required to be implemented in SecureCAN ECUs which want to maintain two values, one from DID and one coded in SW.	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
1	CS_Rx_AccessFailLimit_E	O/M	Boolean	State Encoded

Parameter:
0x00 - Disable
0x01 - Enable

Note: This is parameters enables the overwrite for CS Receive Access Fail limit. This needs to be done only if the supplier maintains two configurations, DID config and software coded config

7.89.14 **DID 0xCB14 Overwrite value for Receive Access Fail Limit**

Service	0x22 / 0x2E	
DID	0xCB14	
Short Description	Manual Overwrite Value for CS Receive Access Fail Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN and DID 0xCB13	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
1	CS_Rx_AccessFailLimit_O	O/M	Unsigned	0x01 - 0xFF

Parameter:
Min - 0x01 (decimal: 1)
Max - 0xFF (decimal: 255)
Default - 0x05 (decimal: 5)

Note: The value defines the authentication failure limit for Receiver ECU

7.89.15 **DID 0xCB15 Overwrite activation for Transmit Authentication Fail Limit**

Service	0x22 / 0x2E	
DID	0xCB15	
Short Description	Manual Overwrite Activation for CS Transmit Authentication Fail Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	This is an optional parameter. It is required to be implemented in SecureCAN ECUs which want to maintain two values, one from DID and one coded in SW.	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
1	CS_Tx_AuthFailLimit_E	O/M	Boolean	State Encoded

Parameter:
0x00 - Disable
0x01 - Enable

Note: This is parameters enables the overwrite for CS Transmit Authentication Fail Limit. This needs to be done only if the supplier maintains two configurations, DID config and software coded config

7.89.16 **DID 0xCB16 Overwrite value for Transmit Authentication Fail Limit**

Service	0x22 / 0x2E	
DID	0xCB16	
Short Description	Manual Overwrite Value for CS Transmit Authentication Fail Limit	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Mandatory for all ECUs which implemented SecureCAN and DID 0xCB15	

Data

Data length (byte)	Parameter Name	Cvt	Type	Value
1	CS_Tx_AuthFailLimit_O	O/M	Unsigned	0x01 - 0xFF

Parameter:
Min - 0x01 (decimal: 1)
Max - 0xFF (decimal: 255)
Default - 0x05 (decimal: 5)

Note: The value defines the authentication failure limit for Sender ECU

7.89.17 **ECU DID 0xCB17 -> 0xCB7F Overwrite for activation of authentication**

Service	0x22 / 0x2E	
DID	0xCB17 -> 0xCB7F	
Short Description	Manual Overwrite for Activation of authentication.	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Optional parameter. This is an ECU specific DID and not System level.	

Data	<table><tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr><tr><td>1</td><td>CS_<Self>_<ECUB>_Auth_E</td><td>O/M</td><td>Boolean</td><td>State Encoded</td></tr></table> **Parameter:** 0x00 - Disable 0x01 - Enable **Note:** This value determines whether the particular authentication needs to be enabled or not. Authentication between Self-ECU and ECU B will need a DID, authentication between Self-ECU and ECU C will need another DID and so on. These DID should be defined in the ECU DID sheet (not System DID sheet) in ODX package.

7.89.18 **ECU DID 0xCB80 -> 0xCBFF Overwrite activation for secret portion verification**

Service	0x22 / 0x2E	
DID	0xCB80 -> 0xCBFF	
Short Description	Manual Overwrite Activation for Secret portion verification of <MSGID>	
Implementation	**Application**	**Flash Bootloader**
	Read/Write	N/A
Security Level	**Read**	**Write**
	Level 2	Level 2
Usage	Optional parameter. This is an ECU specific DID and not System level.	

<table>
<tr>
<td>Data</td>
<td>
<table>
<tr><th>Data length (byte)</th><th>Parameter Name</th><th>Cvt</th><th>Type</th><th>Value</th></tr>
<tr><td>1</td><td>CS_SECCAN_<MSGID>_Auth_E</td><td>O/M</td><td>Boolean</td><td>State Encoded</td></tr>
</table>
Parameter:
0x00 - Disable
0x01 - Enable

Note: The value determines whether the secret portion of the SecureCAN message needs to be verified or not. Each secureCAN message will have a DID like this. These DID should be defined in the ECU DID sheet (not System DID sheet) in ODX package.
</td>
</tr>
</table>

8 Standards

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8.1 Additional Documents

Sr. No	Name	Version
[1]	Basic_Diagnostic_Specification	latest
[2]	Flash_Specification	latest
[3]	Coding_Specification	latest
[4]	ECU_Identifier_Specification	latest
[5]	CAN_Specification	latest
[6]	CyberSecurity	latest

Table 4: Referenced Documents

8.2 Norms and Standards

Ref. No	Document Name

ISO14229	Road vehicles — Unified diagnostic services (UDS)
ISO15031	Road vehicles — Communication between vehicle and external equipment for emissions- related Diagnostic
ISO15765	Road vehicles — Diagnostic on Controller Area Networks (CAN)
ISO17987	Road vehicles — Local Interconnect Network LIN
ISO22900	Road vehicles — Modular vehicle communication interface
ISO11898	Road vehicles – Controlled Area Network (CAN)
ISO22901	Road vehicles — Open diagnostic data exchange (ODX)
SAE J1979	E/E Diagnostic Test Modes
SAE J2012	Diagnostic Trouble Code Definitions

Table 5 : Normative references

This Standard incorporates provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. The respective latest edition of the publication is applicable.

9 Lists

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9.1 List of tables

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9.2 List of Figures

No table of figures entries found.

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Logging configuration

logger.com.intland.codebeamer.export	TRACE
logger.org.docx4j.model.fields.merge	ERROR
logger.velocity	INFO